

Climate results

Loïc Pages

2025-12-17

```
IPM_data <- read.csv("Data/newdata.csv")

MatrixDim <- 50

#Forcer l'age maximal à 8
AgeMax <- 8
IPM_data$Age[IPM_data$Age > AgeMax] <- AgeMax

centauree_data <- IPM_data[!is.na(IPM_data$Size0Mars) & !is.na(IPM_data$Age),]
flower_data <- IPM_data[IPM_data$Flowering==1,] %>% filter(year!=1994)

annees <- 1995:2022
```

Données climatique (température et jours de pluie)

```
climate_data <- read_csv("/media/loic/Commun/0Travail/2025 IPM Centauree ISEM/2025 Stage ISEM/Code/IPM/1")

## Rows: 12053 Columns: 8
## -- Column specification -----
## Delimiter: ","
## db1 (8): AN, MOIS, JOUR, precipitations, T moyenne, T min,, T max,, Prec > 1 mm
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

climate_data <- climate_data %>%
  rename(
    "Tmean" = "T moyenne",
    "Tmin" = "T min,",
    "Tmax" = "T max,",
    "preci1mm" = "Prec > 1 mm",
    "year" = "AN"
  ) %>%
  filter(!year<1994)

# Mean for each season of each year
climate_data_season <- climate_data %>%
  mutate(
    year_season = if_else(MOIS == 12, year + 1L, year),
    Season = case_when(
      MOIS %in% c(12, 1, 2) ~ "Winter",
```

```

    MOIS %in% c(3, 4, 5) ~ "Spring",
    MOIS %in% c(6, 7, 8) ~ "Summer",
    MOIS %in% c(9, 10, 11) ~ "Autumn")) %>%
group_by(year = year_season, Season) %>%
summarise(
  Tmean      = mean(Tmean, na.rm = TRUE),
  NbWetDays  = sum(precipmm, na.rm = TRUE),
  Tmax       = max(Tmax, na.rm = TRUE),
  .groups    = "drop") %>%
filter(year!=2023)

# Mean over administrative year
climate_data_mean <- climate_data %>%
  group_by(year) %>%
  summarise(
    Tmean      = mean(Tmean, na.rm = TRUE),
    NbWetDays  = sum(precipmm, na.rm = TRUE),
    Tmax       = max(Tmax, na.rm = TRUE)) %>%
  filter(year!=1994)

# climate_data_month <- climate_data %>%
#   mutate(MOIS = as.numeric(MOIS)) %>%
#   group_by(year, MOIS) %>%
#   summarise(Tmean = mean(Tmean),
#             NbWetDays = sum(precipmm),
#             Tmax = max(Tmax))
#
# for (i in 1:5){
#   climate_data_month$year[climate_data_month$MOIS==i] <- climate_data_month[climate_data_month$MOIS==i,
# }

```

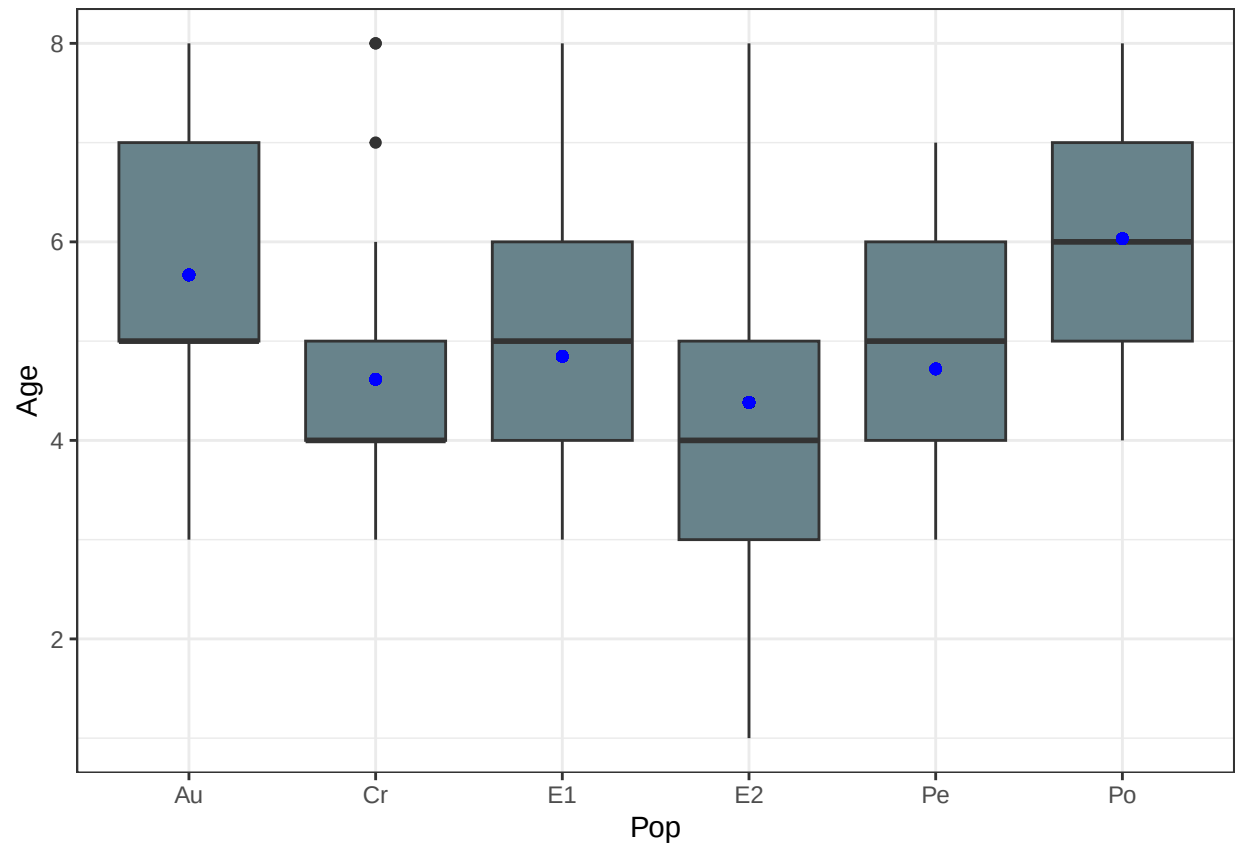
Observed data

Observed mean age at flowering

```

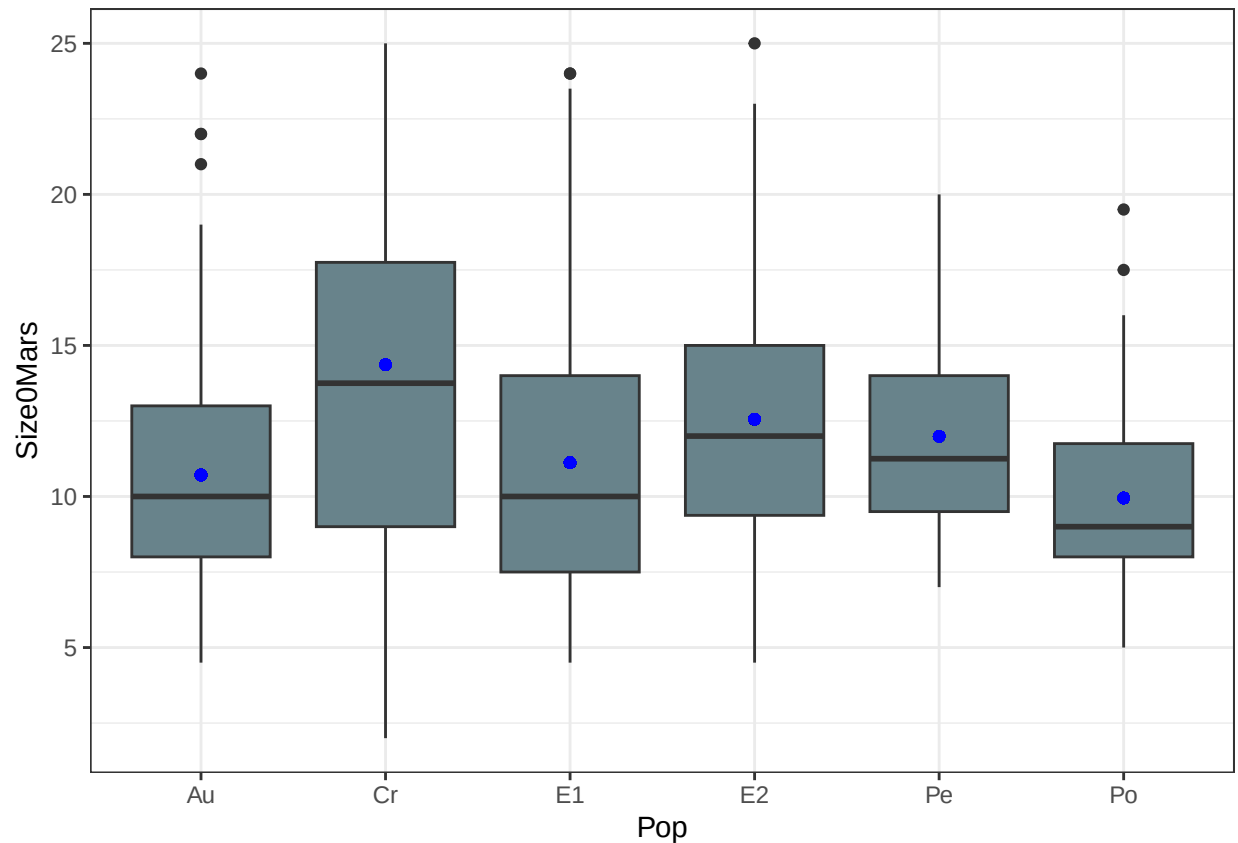
flower_data %>%
  group_by(Pop) %>%
  mutate(mean.age = mean(Age, na.rm=TRUE)) %>%
  ggplot(aes(x=Pop, y=Age)) +
  geom_boxplot(fill="lightblue4")+
  geom_point(aes(y=mean.age), color= "blue")+
  theme_bw()

```



Observed mean size at flowering

```
flower_data %>%
  group_by(Pop) %>%
  mutate(mean.size = mean(Size0Mars,na.rm=TRUE)) %>%
  ggplot(aes(x=Pop, y=Size0Mars)) +
  geom_boxplot(fill="lightblue4")+
  geom_point(aes(y=mean.size),color= "blue")+
  theme_bw()
```

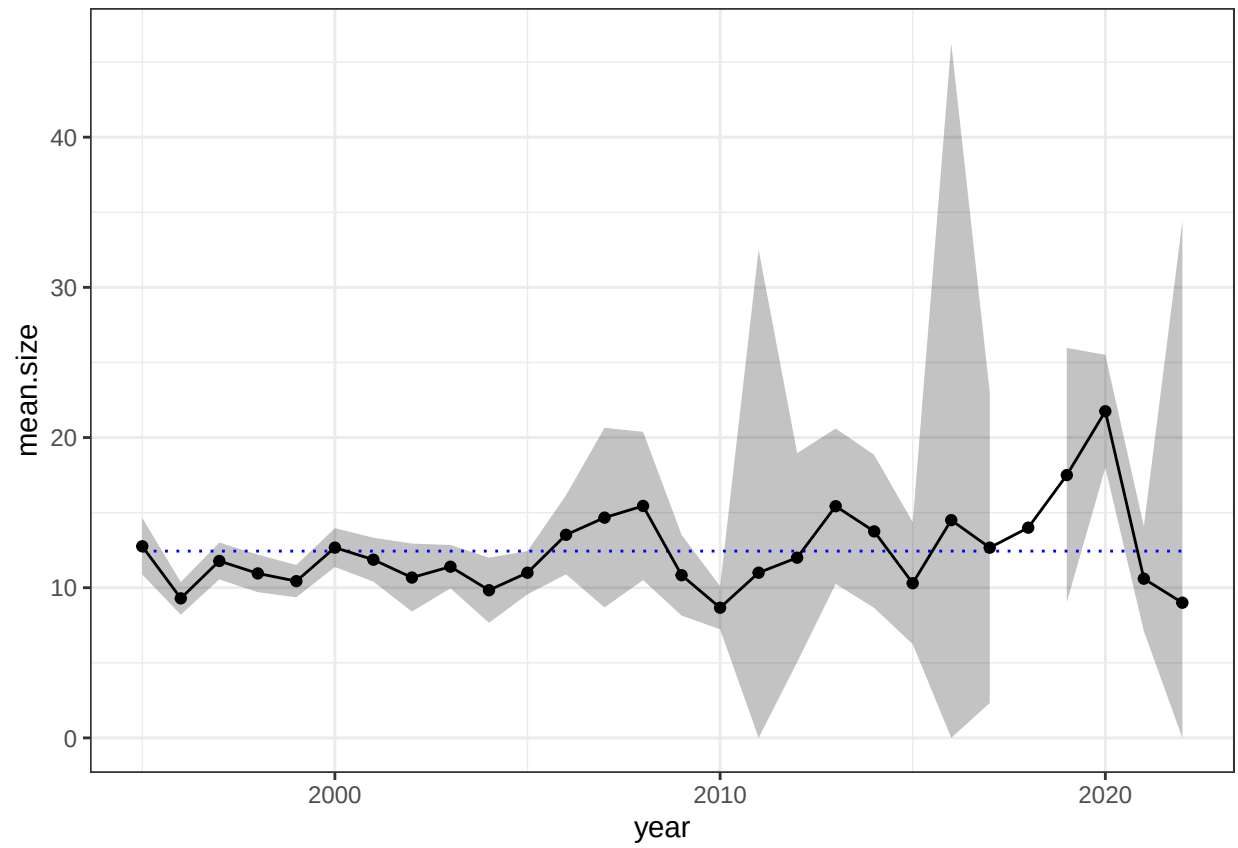


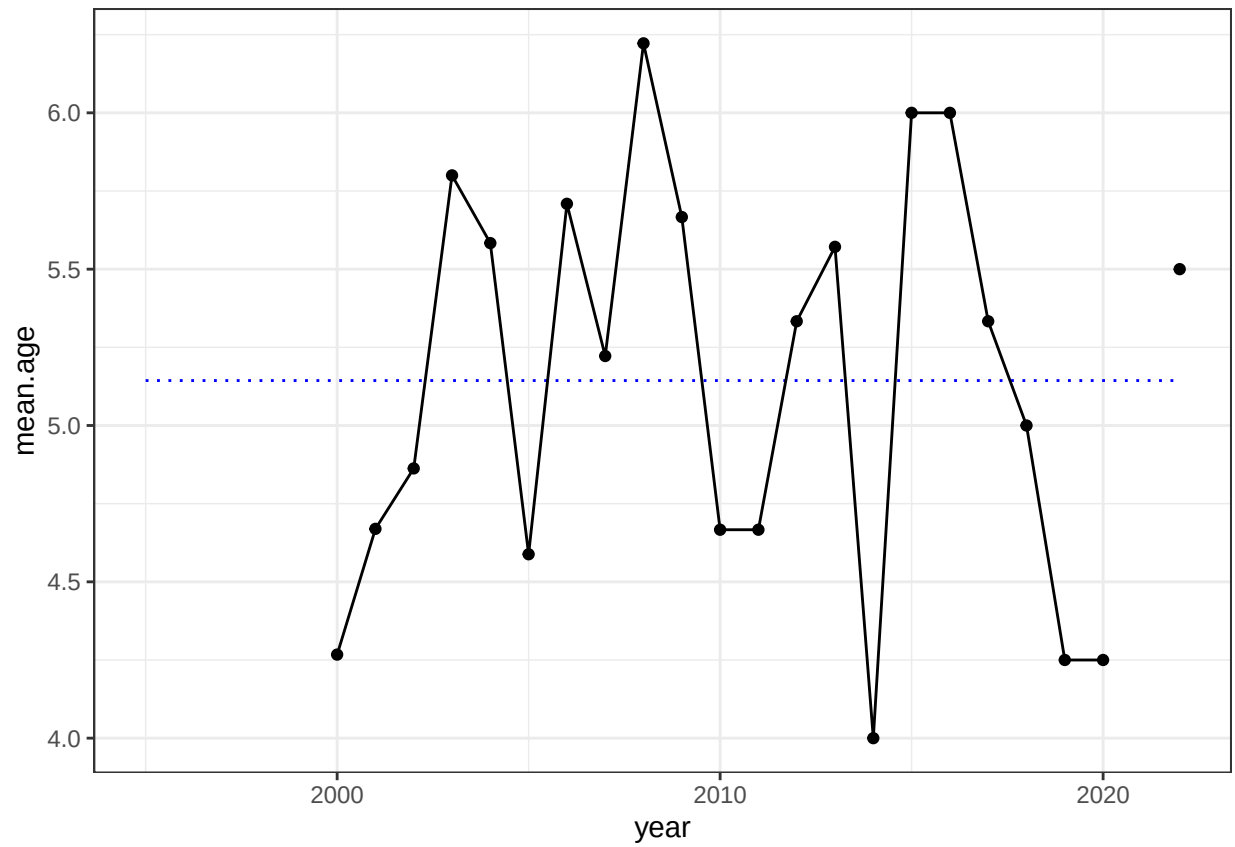
On calcule la taille et l'âge moyen annuel des plantes en fleur, pondéré par le nombre d'individus dans chaque population. On calcule aussi l'intervalle de confiance de la taille moyenne.

```
pop_stats <- flower_data %>%
  group_by(year, Pop) %>%
  summarise( mean.size_pop = mean(Size0Mars, na.rm = TRUE),
             sd.size_pop = sd(Size0Mars, na.rm=TRUE),
             mean.age_pop = mean(Age, na.rm = TRUE),
             n_pop = n(),
             .groups = "drop" )

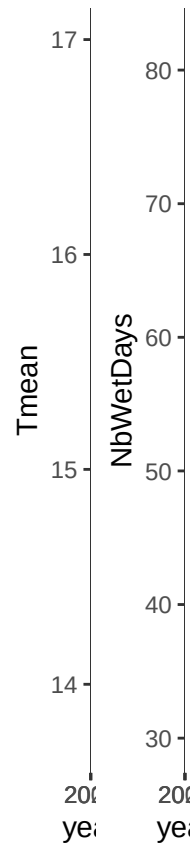
flow_data_mean <- pop_stats %>%
  group_by(year) %>%
  summarise(
    mean.size = sum(mean.size_pop * n_pop) / sum(n_pop),
    mean.age = sum(mean.age_pop * n_pop) / sum(n_pop),
    nbrIndyear = sum(n_pop),
    # variance combinée
    var.size = (sum((n_pop - 1)*sd.size_pop^2, na.rm = TRUE) + sum(n_pop*(mean.size_pop-sum(mean.size_p
    / (sum(n_pop) - 1),
    .groups = "drop") %>%
  mutate(sd.size = sqrt(var.size),
         se.size = sd.size / sqrt(nbrIndyear),
         # IC à 95 %
         ci.low = mean.size - qt(0.975, df = nbrIndyear - 1) * se.size,
         ci.high = mean.size + qt(0.975, df = nbrIndyear - 1) * se.size)
```

```
flow_data_mean <- flow_data_mean %>% mutate(ci.low = ifelse(ci.low<0,0,ci.low))
```





L'intervalle de confiance de la taille moyenne devient très grand sur la fin à cause du très faible nombre de plantes en fleur.

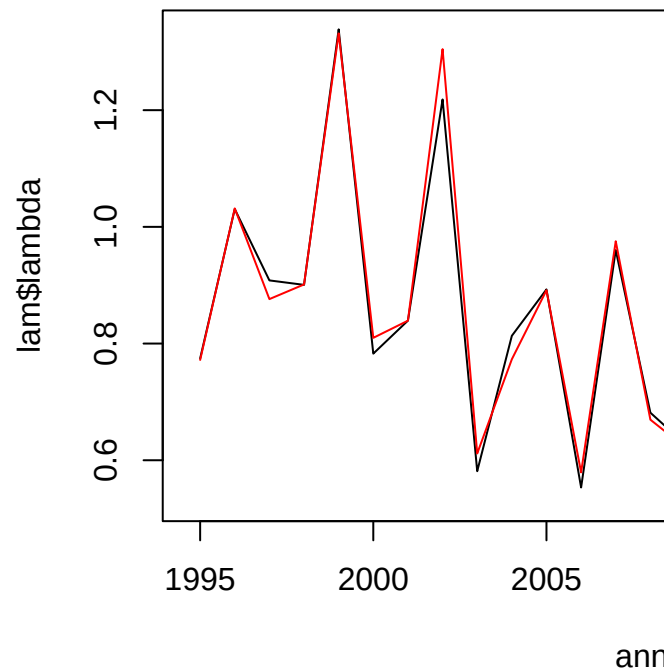


Température et nombre de jours de pluie annuel.

Growth rate

```
load("IPM/Results/lambdaYear.RData")
load("IPM/Results/lam_opti_year.RData")
load("IPM/Results/lam_opti_popyear.RData")
```

```
for(i in 1:length(annees)){
  lam_opti_year[i] <- mean(lam_opti_popyear[,i])}
```



Optimal growth rate (red) compared to predicted growth red (black)

Sensitivity to climatic window

```
seasons <- c("Spring", "Summer", "Autumn", "Winter")

# Sensibilité aux saisons de l'année t
for(s in seasons){
test_clim <- climate_data_season %>%
  ungroup() %>%
  filter(Season==s,
         year!=1994) %>%
  mutate(lambda = lam_opti_year)

print(paste(s, "at t, r²=", round(summary(lm(lambda~Tmean, data=(test_clim)))$r.squared, 5)))
}
```

```
## [1] "Spring at t, r²= 0.00518"
## [1] "Summer at t, r²= 0.07527"
## [1] "Autumn at t, r²= 0.03972"
## [1] "Winter at t, r²= 0.01745"
```

```
# Sensibilité aux saisons de l'année t-1
for(s in seasons){
test_clim <- climate_data_season %>%
```

```

ungroup() %>%
filter(Season==s,
       year!=2022) %>%
mutate(year = year+1,
       lambda = lam_opti_year)

print(paste(s,"at t-1, r²=", round(summary(lm(lambda~Tmean,data=(test_clim)))$r.squared,5)))
}

```

```

## [1] "Spring at t-1, r²= 0.04003"
## [1] "Summer at t-1, r²= 0.01605"
## [1] "Autumn at t-1, r²= 0.00281"
## [1] "Winter at t-1, r²= 0.00277"

```

```

test_clim <- climate_data_season %>%
  ungroup() %>%
  filter(Season %in% c("Summer","Autumn"),
         year!=1994) %>%
  group_by(year) %>%
  summarise(Tmean = mean(Tmean)) %>%
  ungroup() %>%
  mutate(lambda = lam_opti_year)

print(paste("Summer and Autumn","at t, r²=", round(summary(lm(lambda~Tmean,data=(test_clim)))$r.squared

```

```

## [1] "Summer and Autumn at t, r²= 0.08174"

```

```

test_clim <- climate_data_season %>%
  ungroup() %>%
  filter(Season %in% c("Summer","Spring"),
         year!=1994) %>%
  group_by(year) %>%
  summarise(Tmean = mean(Tmean)) %>%
  ungroup() %>%
  mutate(lambda = lam_opti_year)

print(paste("Summer and Spring","at t, r²=", round(summary(lm(lambda~Tmean,data=(test_clim)))$r.squared

```

```

## [1] "Summer and Spring at t, r²= 0.05798"

```

```

test_clim <- climate_data_season %>%
  ungroup() %>%
  filter(Season %in% c("Winter","Spring"),
         year!=1994) %>%
  group_by(year) %>%
  summarise(Tmean = mean(Tmean)) %>%
  ungroup() %>%
  mutate(lambda = lam_opti_year)

print(paste("Winter and Spring","at t, r²=", round(summary(lm(lambda~Tmean,data=(test_clim)))$r.squared

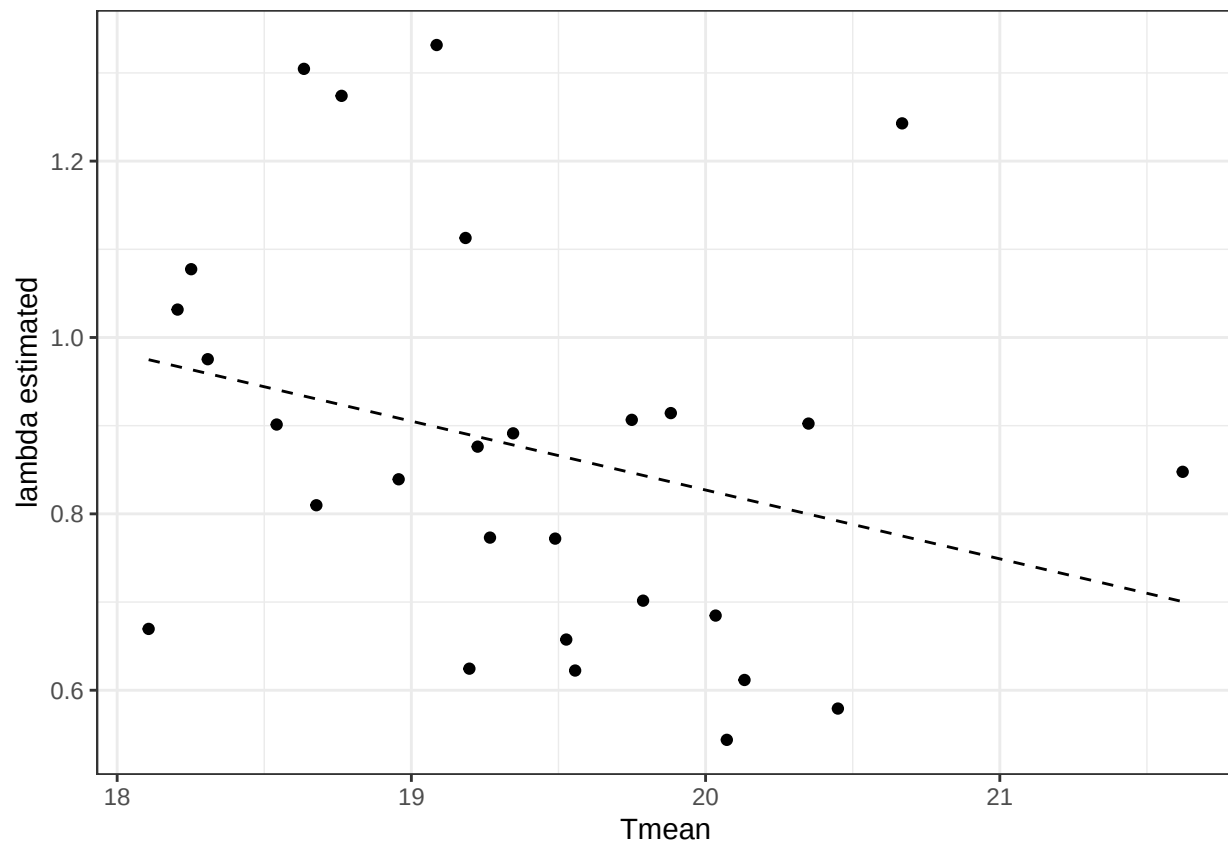
```

```
## [1] "Winter and Spring at t, r²= 0.01678"
```

Variations to climatic parameters

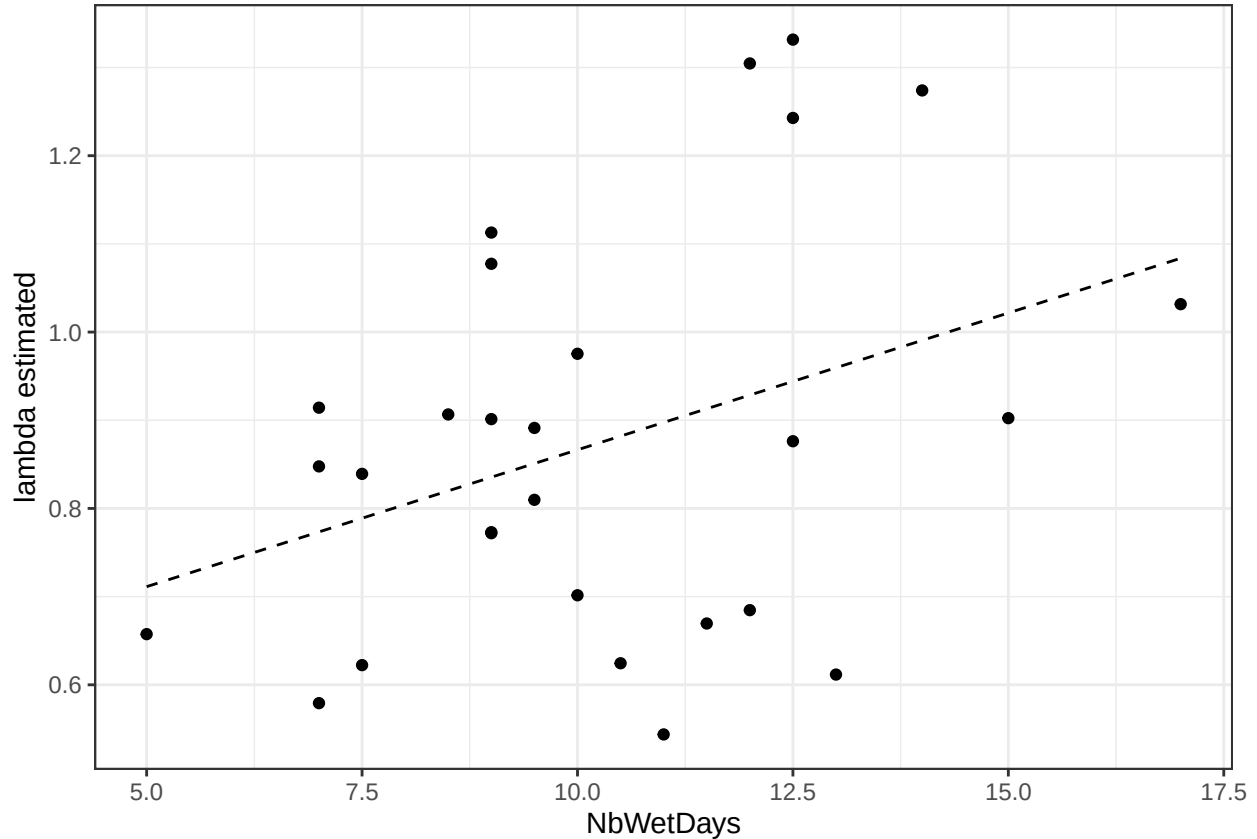
```
climate_data_mean_lambda <- climate_data_season %>%  
  ungroup() %>%  
  filter(Season %in% c("Summer", "Autumn"),  
         year!=1994) %>%  
  group_by(year) %>%  
  summarise(Tmean = mean(Tmean),  
            NbWetDays = mean(NbWetDays)) %>%  
  ungroup() %>%  
  mutate(lambda = lam_opti_year)
```

```
climate_data_mean_lambda %>%  
  ggplot(aes(x=Tmean,y=lambda)) +  
  geom_point()+  
  geom_line(aes(y=predict(lm(lambda~Tmean,data=(climate_data_mean_lambda )))),linetype=2)+  
  theme_bw()+  
  ylab("lambda estimated")
```



```
climate_data_mean_lambda %>%  
  ggplot(aes(x=NbWetDays,y=lambda)) +
```

```
geom_point()+
geom_line(aes(y=predict(lm(lambda~NbWetDays,data=(climate_data_mean_lambda)))),linetype=2)+
theme_bw()+
ylab("lambda estimated")
```



```
summary(lm(lambda~Tmean,data=climate_data_mean_lambda))$r.squared
```

```
## [1] 0.08173866
```

```
summary(lm(lambda~NbWetDays,data=climate_data_mean_lambda))$r.squared
```

```
## [1] 0.1373664
```

Conclusion

Le taux de croissance (lambda) optimal est très proche du lambda prédit par le modèle.

Le lambda optimal à l'année t est plus sensible aux variations de températures au printemps-été de l'année t. Il est négativement corrélé à la température moyenne durant cet intervalle de temps, et positivement corrélé avec le nombre de jours de pluie durant cet intervalle de temps.

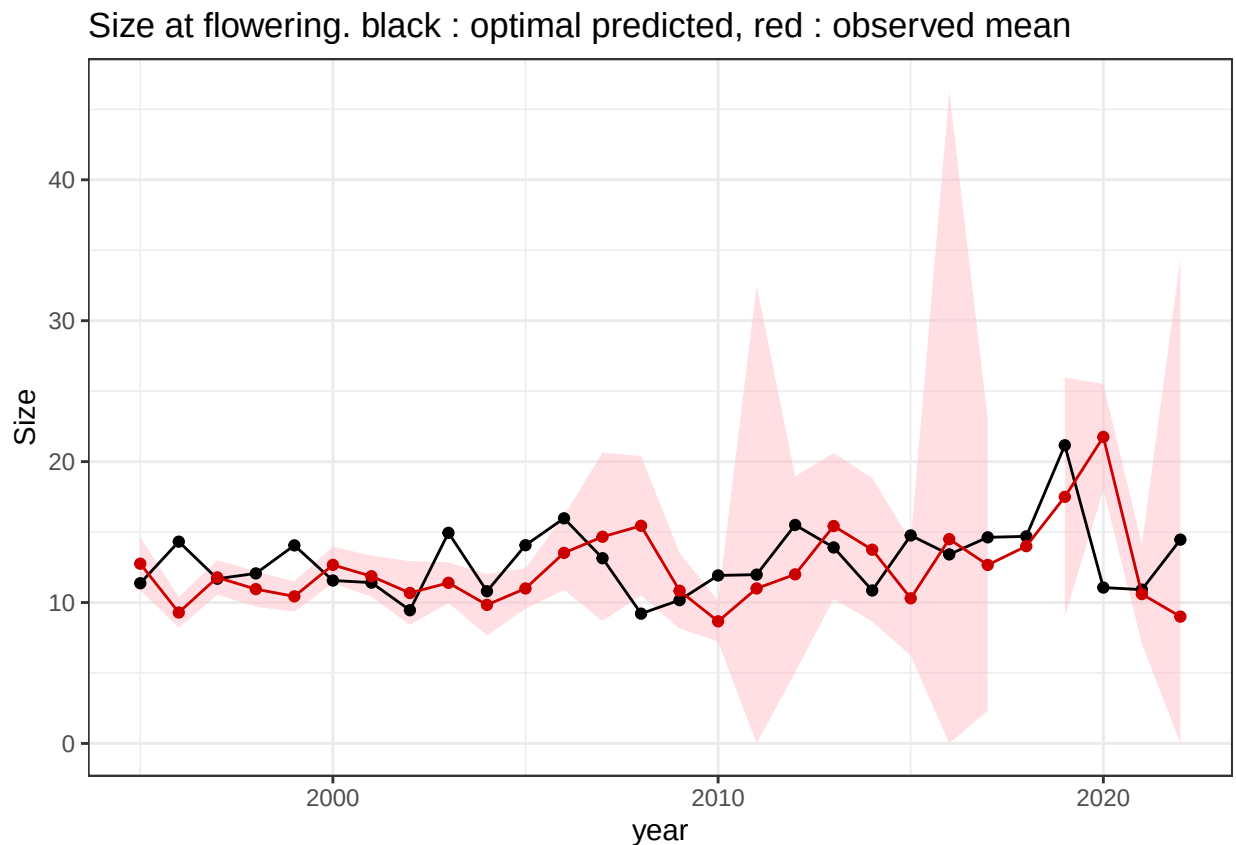
Size at flowering

```
for(i in 1:length(annees)){  
  opti_size_year[i] <- mean(opti_size_popyear[,i])  
}
```

Comparison of optimal size to mean observed size

Optimal size at flowering for each year - the model consider that for each year, if we stay in this year indefinitely, the optimal flowering size is opti.size.

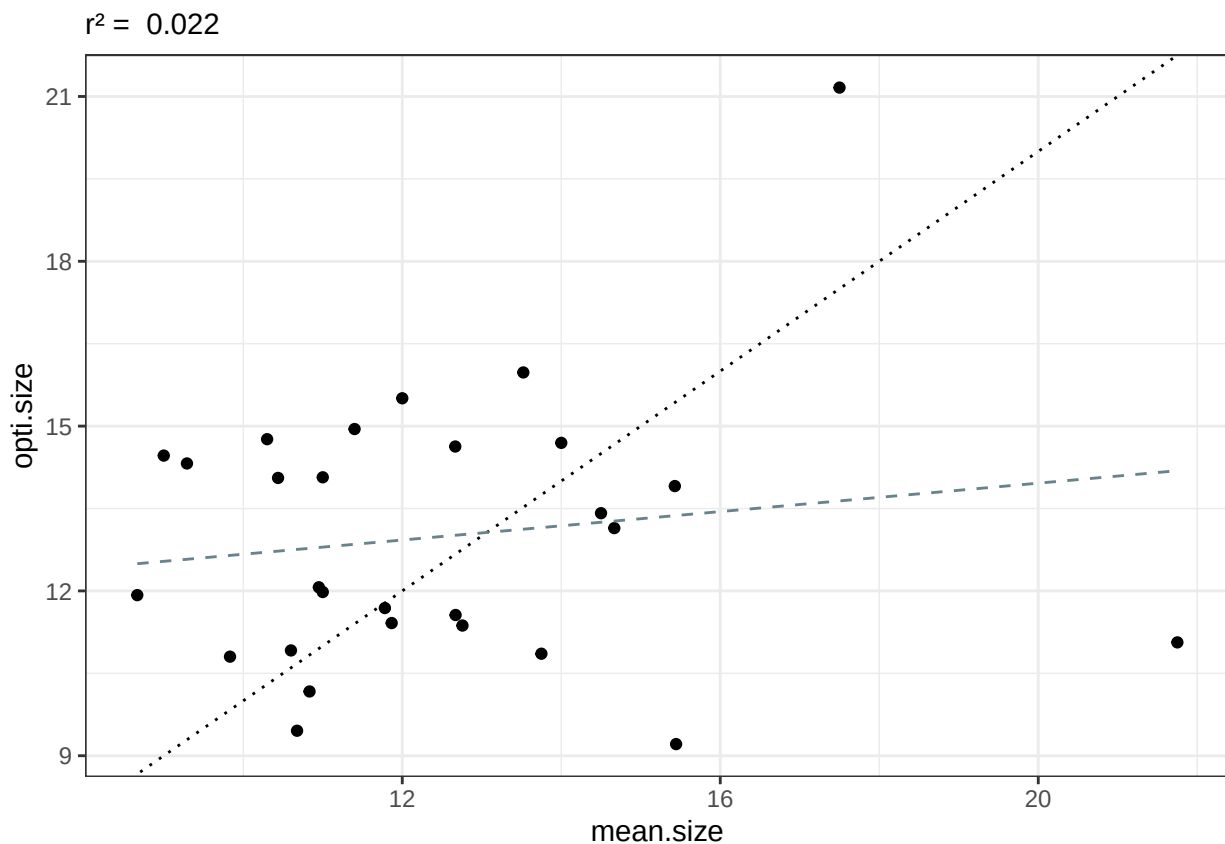
```
optimalsize_data %>%  
  ggplot(aes(x=year)) +  
  geom_ribbon(aes(ymax=ci.high,ymin=ci.low),fill="pink",alpha=0.5)+  
  geom_point(aes(y=opti.size))+  
  geom_line(aes(y=opti.size))+  
  geom_point(aes(y=mean.size),col="red3")+  
  geom_line(aes(y=mean.size),col="red3")+  
  # geom_point(aes(y=predict.size),col="blue")+  
  # geom_line(aes(y=predict.size),col="blue")+  
  theme_bw()+  
  labs(title="Size at flowering. black : optimal predicted, red : observed mean")+  
  ylab("Size")
```



Optimal size at flowering compared to observed mean size at flowering

```
r2 <- optimalsize_data %>%
  mutate(lm = predict(lm(opti.size~mean.size))) %>%
  summarize(rsquared = summary(lm(opti.size~mean.size))$r.squared)

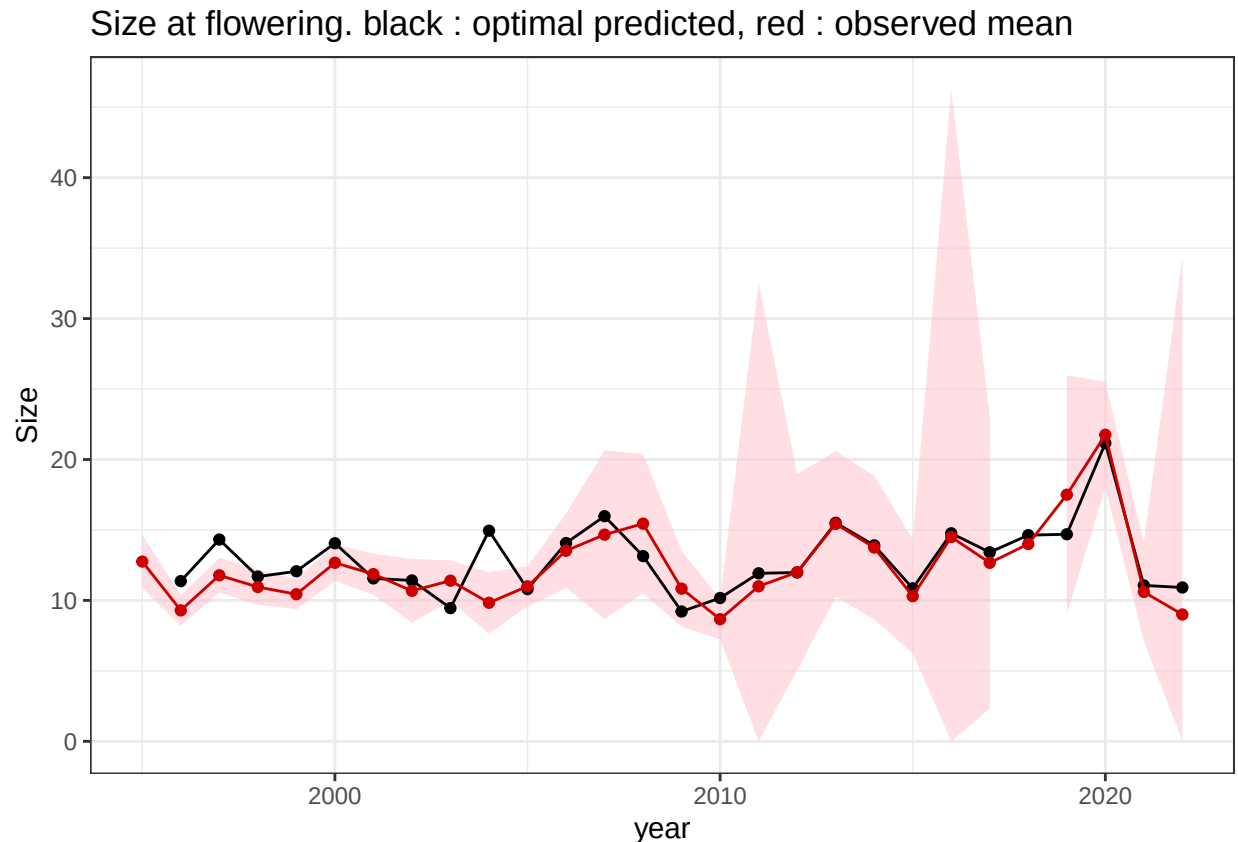
optimalsize_data %>%
  mutate(lm = predict(lm(opti.size~mean.size))) %>%
  ggplot(aes(x=mean.size, y=opti.size)) +
  geom_point()+
  geom_abline(intercept=0,slope=1,linetype=3)+
  geom_line(aes(y=lm), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(subtitle = str_c("r2 = ",round(r2,3)))
```



Optimal size at flowering for each year If we assume that it is not the current year but the previous year which impact the choice of flowering, so we compare the mean size at flowering at year t to the optimal size at year t-1.

```
optimalsize_data %>%
  mutate(opti.size = c(NA,opti.size[-28])) %>%
  ggplot(aes(x=year)) +
  geom_ribbon(aes(ymax=ci.high,ymin=ci.low),fill="pink",alpha=0.5)+
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=opti.size))+
  geom_point(aes(y=mean.size),col="red3")+
  theme_bw()
```

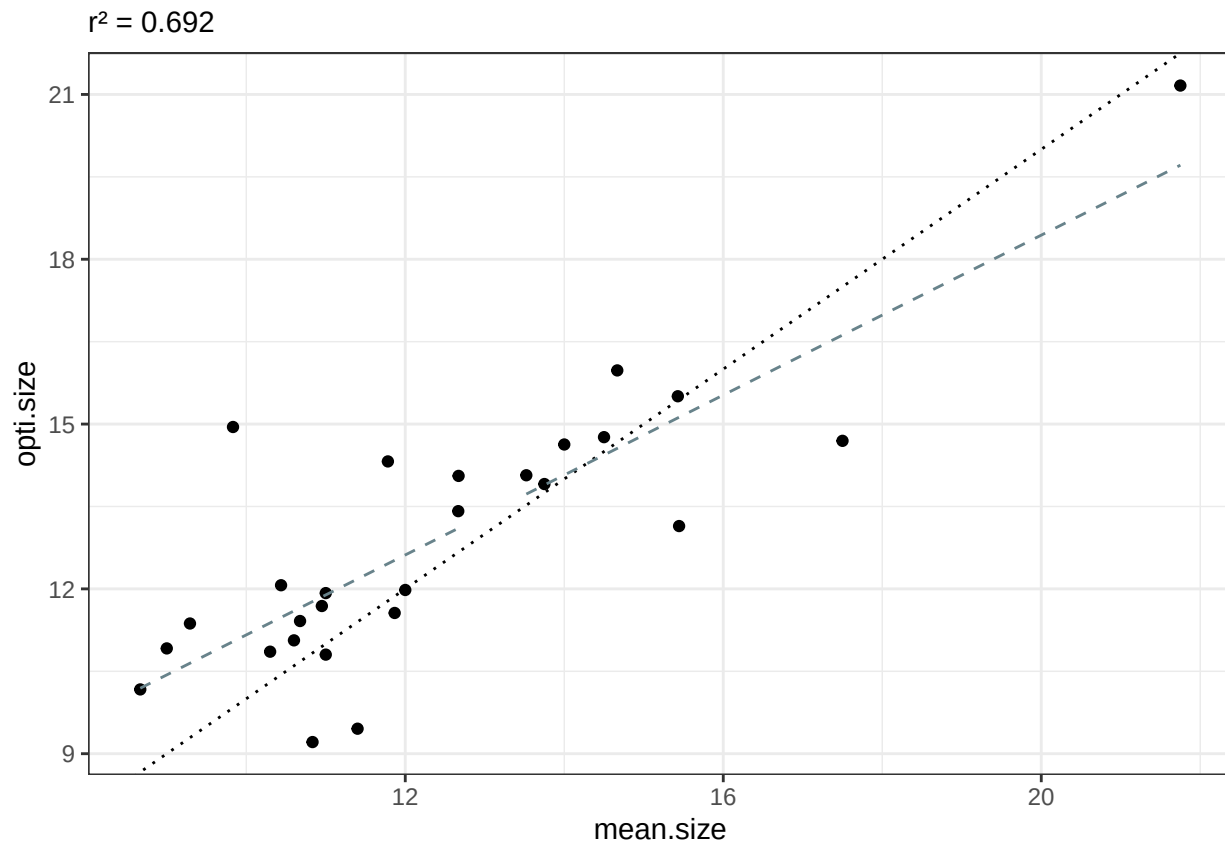
```
geom_line(aes(y=mean.size),col="red3")+
# geom_point(aes(y=predict.size),col="blue")+
# geom_line(aes(y=predict.size),col="blue")+
theme_bw()+
labs(title="Size at flowering. black : optimal predicted, red : observed mean")+
ylab("Size")
```



Optimal size at flowering compared to observed mean size at flowering

```
r2 <- optimalsize_data %>%
  mutate(opti.size = c(NA,opti.size[-28]),
         lm = c(NA,predict(lm(opti.size~mean.size)))) %>%
  summarize(rsquared = summary(lm(opti.size~mean.size))$r.squared)

optimalsize_data %>%
  mutate(opti.size = c(NA,opti.size[-28]),
         lm = c(NA,predict(lm(opti.size~mean.size))),
         rsquared = rep(summary(lm(opti.size~mean.size))$r.squared,28)) %>%
  ggplot(aes(x=mean.size, y=opti.size)) +
  geom_point()+
  geom_abline(intercept=0,slope=1,linetype=3)+
  geom_line(aes(y=lm), linetype=2, col="lightblue4")+
  theme_bw()+
  labs(subtitle = str_c("r2 = ",as.character(round(r2,3))))
```



Sensitivity to climatic window

```
seasons <- c("Spring", "Summer", "Autumn", "Winter")

# Sensibilité aux saisons de l'année t
for(s in seasons){
  test_clim <- climate_data_season %>%
    ungroup() %>%
    filter(Season==s,
           year!=1994) %>%
    mutate(opti.size = opti_size_year)

  print(paste(s, "at t, r²=", round(summary(lm(opti.size~Tmean, data=(test_clim)))$r.squared, 5)))
}
```

```
## [1] "Spring at t, r²= 0.00954"
## [1] "Summer at t, r²= 0.19465"
## [1] "Autumn at t, r²= 0.02972"
## [1] "Winter at t, r²= 0.01543"
```

```
# Sensibilité aux saisons de l'année t-1
for(s in seasons){
  test_clim <- climate_data_season %>%
```

```

ungroup() %>%
filter(Season==s,
      year!=2022) %>%
mutate(year = year+1,
      opti.size = opti_size_year)

print(paste(s,"at t-1, r²=", round(summary(lm(opti.size~Tmean,data=(test_clim)))$r.squared,5)))
}

```

```

## [1] "Spring at t-1, r²= 0.00066"
## [1] "Summer at t-1, r²= 0.03757"
## [1] "Autumn at t-1, r²= 0.22562"
## [1] "Winter at t-1, r²= 0.00932"

```

```

test_clim <- climate_data_season %>%
ungroup() %>%
filter(Season %in% c("Autumn","Summer"),
      year!=1994) %>%
group_by(year) %>%
summarise(Tmean = mean(Tmean)) %>%
ungroup() %>%
mutate(opti.size = opti_size_year)

print(paste("Summer and Autumn","at t, r²=", round(summary(lm(opti.size~Tmean,data=(test_clim)))$r.squared,5)))

```

```

## [1] "Summer and Autumn at t, r²= 0.13562"

```

```

test_clim <- climate_data_season %>%
ungroup() %>%
filter(Season %in% c("Summer","Spring"),
      year!=1994) %>%
group_by(year) %>%
summarise(Tmean = mean(Tmean)) %>%
ungroup() %>%
mutate(opti.size = opti_size_year)

print(paste("Summer and Spring","at t, r²=", round(summary(lm(opti.size~Tmean,data=(test_clim)))$r.squared,5)))

```

```

## [1] "Summer and Spring at t, r²= 0.14177"

```

```

test_clim <- climate_data_season %>%
ungroup() %>%
filter(Season %in% c("Autumn","Summer"),
      year!=2022) %>%
group_by(year) %>%
summarise(Tmean = mean(Tmean)) %>%
ungroup() %>%
mutate(year = year+1,
      opti.size = opti_size_year)

print(paste("Autumn and Summer at t-1, r²=", round(summary(lm(opti.size~Tmean,data=(test_clim)))$r.squared,5)))

```

```
## [1] "Autumn and Summer at t-1, r2= 0.18919"
```

```
test_clim <- climate_data_season %>%
  ungroup() %>%
  filter(Season %in% c("Autumn", "Winter"),
         year!=2022) %>%
  group_by(year) %>%
  summarise(Tmean = mean(Tmean)) %>%
  ungroup() %>%
  mutate(year = year+1,
         opti.size = opti_size_year)

print(paste("Autumn and Winter at t-1, r2=", round(summary(lm(opti.size~Tmean,data=(test_clim)))$r.squa
```

```
## [1] "Autumn and Winter at t-1, r2= 0.04692"
```

```
test_clim <- climate_data_season %>%
  ungroup() %>%
  filter(Season %in% c("Autumn", "Winter"),
         year!=2022) %>%
  group_by(year) %>%
  summarise(Tmean = mean(Tmean)) %>%
  ungroup() %>%
  mutate(year = year+1,
         opti.size = opti_size_year)

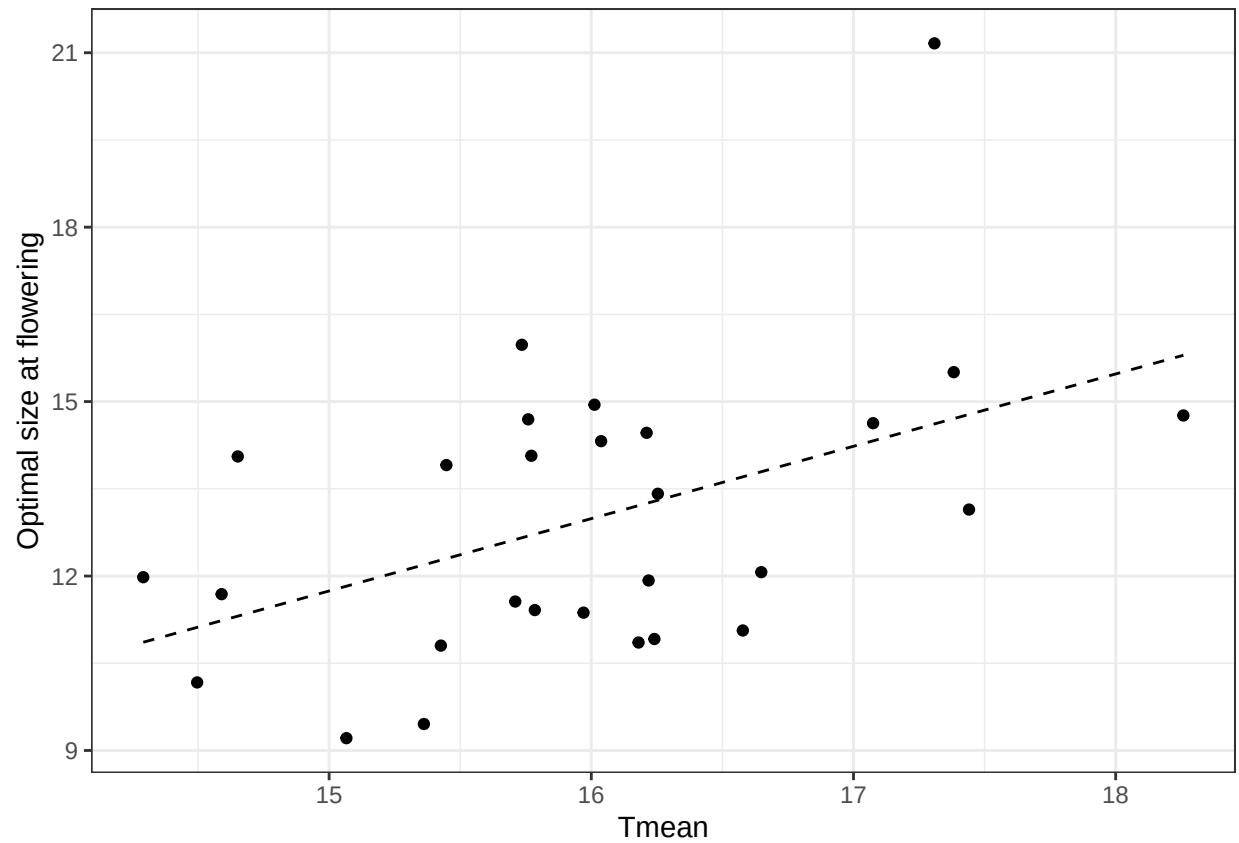
print(paste("Autumn and Winter at t-1, r2=", round(summary(lm(opti.size~Tmean,data=(test_clim)))$r.squa
```

```
## [1] "Autumn and Winter at t-1, r2= 0.04692"
```

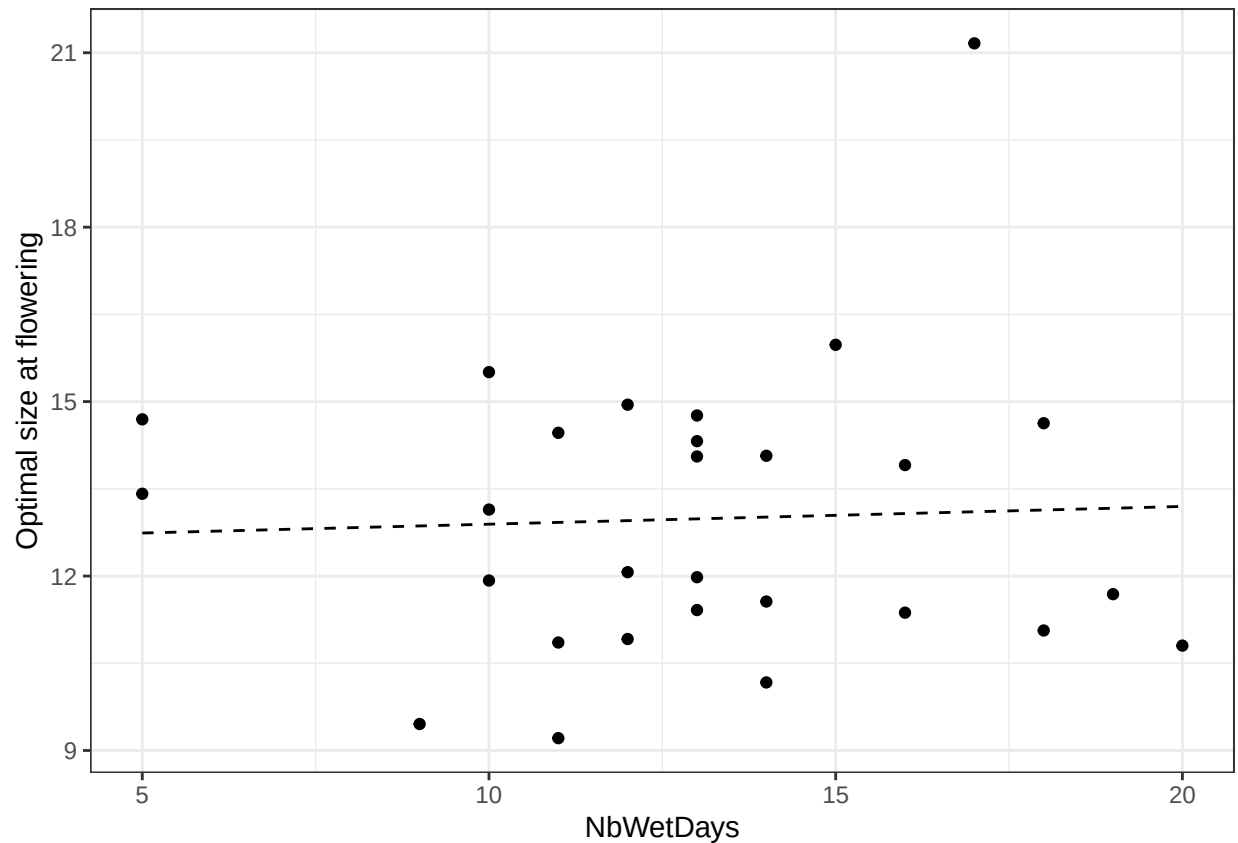
Variations to climatic parameters

```
climate_data_mean_size <- climate_data_season %>%
  ungroup() %>%
  filter(Season=="Autumn",
         year!=2022) %>%
  mutate(year = year+1,
         opti.size = opti_size_year)
```

```
climate_data_mean_size %>%
  ggplot(aes(x=Tmean,y=opti.size)) +
  geom_point()+
  geom_line(aes(y=predict(lm(opti.size~Tmean,data=(climate_data_mean_size )))),linetype=2)+
  theme_bw()+
  ylab("Optimal size at flowering")
```



```
climate_data_mean_size %>%  
  ggplot(aes(x=NbWetDays,y=opti.size)) +  
  geom_point()+  
  geom_line(aes(y=predict(lm(opti.size~NbWetDays,data=(climate_data_mean_size)))),linetype=2)+  
  theme_bw()+  
  ylab("Optimal size at flowering")
```



```
summary(lm(opti.size~Tmean,data=climate_data_mean_size))$r.squared
```

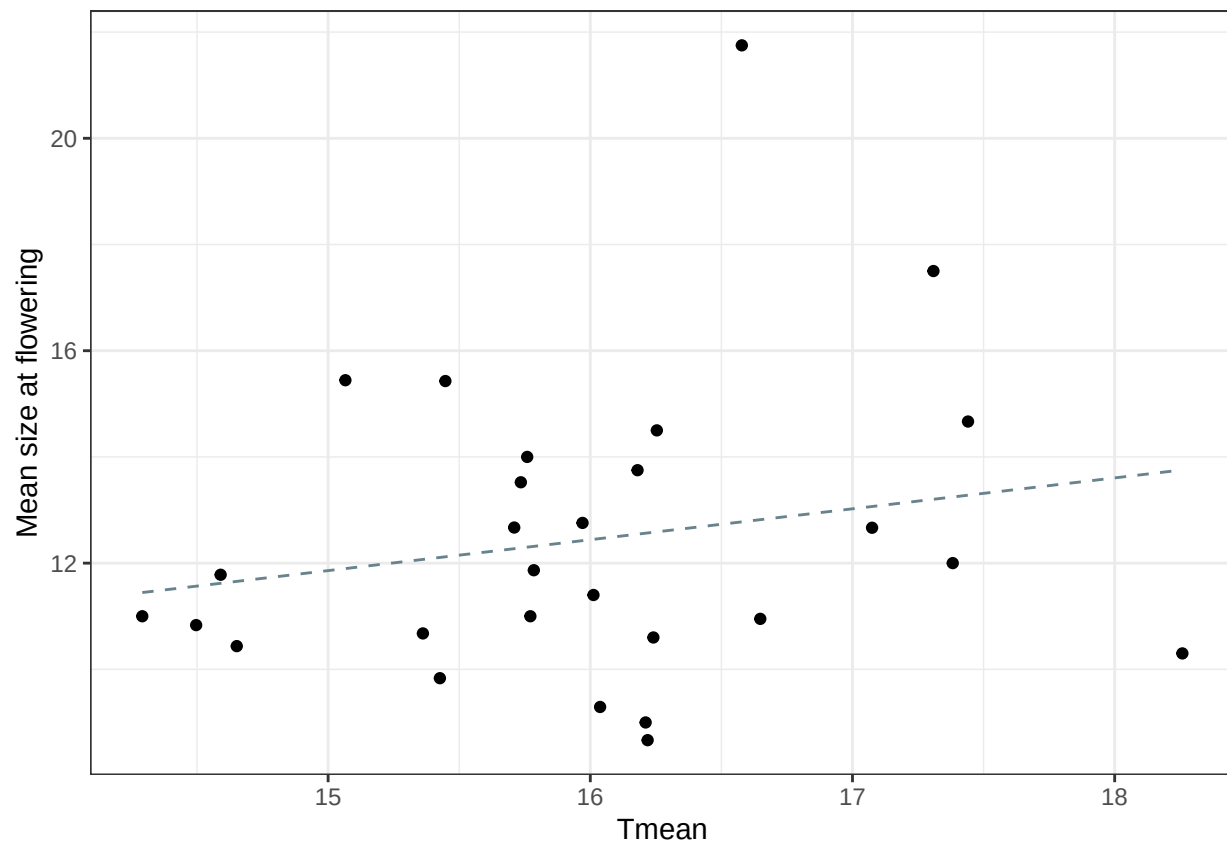
```
## [1] 0.2256221
```

```
summary(lm(opti.size~NbWetDays,data=climate_data_mean_size ))$r.squared
```

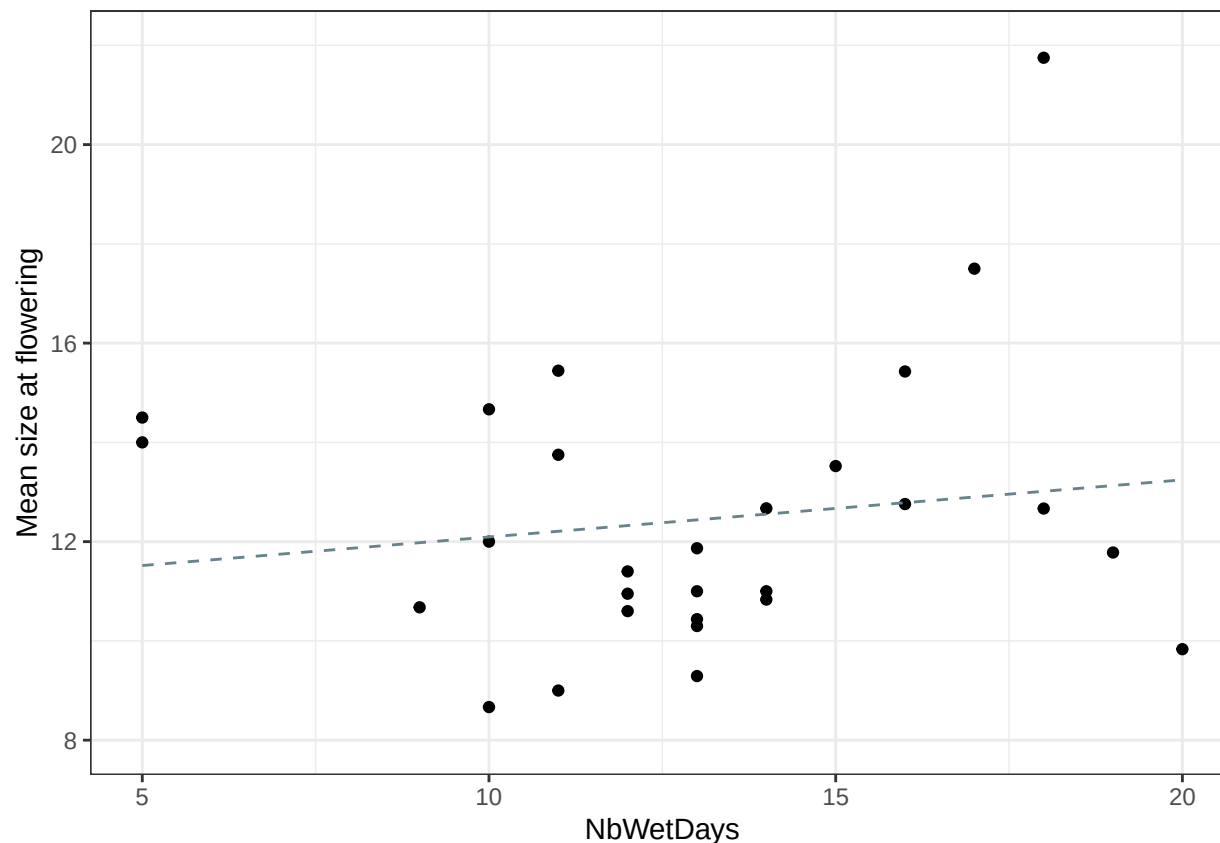
```
## [1] 0.002028336
```

```
flowclimate_data <- left_join(climate_data_mean_size, optimalsize_data, by="year")
```

```
flowclimate_data %>%
  ggplot(aes(x=Tmean)) +
  geom_point(aes(y=mean.size)) +
  geom_line(aes(y=predict(lm(mean.size~Tmean,data=flowclimate_data))), linetype=2, col="lightblue4") +
  theme_bw() +
  labs(y="Mean size at flowering")
```



```
flowclimate_data %>%
  ggplot(aes(x=NbWetDays)) +
  geom_point(aes(y=mean.size))+
  geom_line(aes(y=predict(lm(mean.size~NbWetDays,data=flowclimate_data))), linetype=2, col="lightblue4",
  theme_bw()+
  labs(y="Mean size at flowering")+
  ylim(8,22)
```



```
summary(lm(mean.size~Tmean,data=flowclimate_data))$r.squared
```

```
## [1] 0.03828487
```

```
summary(lm(mean.size~NbWetDays,data=flowclimate_data))$r.squared
```

```
## [1] 0.02242854
```

Conclusion

La taille optimale à la floraison semble décalée de la taille à la floraison observée. Les effets de l'année t semblent affecter les tailles à la floraison de l'année $t+1$. Après avoir fait ce décalage, les tailles optimales sont très proches des tailles observées. Ce qui pourrait s'expliquer par le fait que la variation du β sur la fonction de floraison n'augmente pas vraiment le taux de croissance, et donc que la taille à la floraison est déjà proche de l'optimum.

On confirme cette hypothèse en constatant que la taille optimale à la floraison est plus sensible à la température de l'automne de l'année $t-1$ que des températures à l'année t .

la taille optimale à la floraison est positivement corrélée à la température moyenne en automne à $t-1$, mais n'est pas corrélée au nombre de jours de pluie.

Comparison with other life history traits

```
flowclimate_data <- left_join(climate_data_mean,optimalsize_data,by="year") %>% filter(year!=1994)

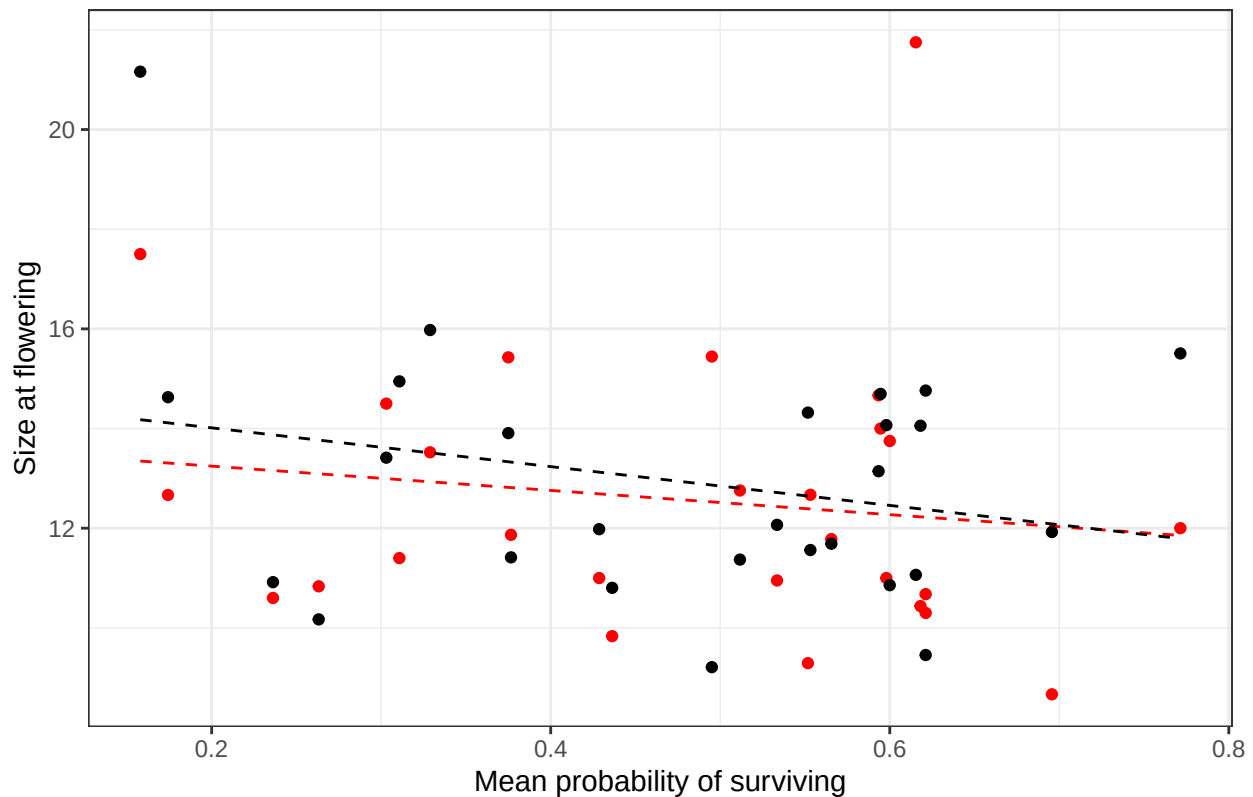
flowclimate_data2 <- flowclimate_data %>%
  left_join(IPM_data %>%
    group_by(year) %>%
    summarize(mean.surv = mean(Survie))) %>%
  left_join(flower_data %>%
    group_by(year) %>%
    summarize(mean.cap = mean(ifelse(is.na(Capitule), exp(2.31070+0.06846*SizeOMars), Capitule))) %>%
  left_join(IPM_data %>%
    filter(Age==1) %>%
    group_by(year) %>%
    summarize(Seedlings = sum(Age)))

## Joining with 'by = join_by(year)'
## Joining with 'by = join_by(year)'
## Joining with 'by = join_by(year)'

#
# cctl <- flower_data %>%
#   mutate(Capitule = ifelse(is.na(Capitule), exp(2.31070+0.06846*SizeOMars), Capitule)) %>%
#   group_by(Quadrat,year,Pop) %>%
#   summarize(Capitule = sum(Capitule))
# # cctl$year <- cctl$year+1
# plt <- IPM_data %>%
#   filter(Age==1) %>%
#   group_by(Quadrat,year,Pop) %>%
#   summarize(Seedlings = sum(Age))
# Estb <- left_join(plt, cctl, by = join_by(Quadrat, year, Pop))
#
# flowclimate_data2 <- flowclimate_data2 %>%
#   left_join(Estb %>%
#     group_by(year) %>%
#     summarize(mean.seedling = mean(Seedlings,na.rm=TRUE),
#               mean.cap = mean(Capitule,na.rm=TRUE)))

flowclimate_data2 %>%
  filter(year!=2022) %>%
  ggplot(aes(x=mean.surv))+
  geom_point(aes(y=mean.size),col="red")+
  geom_line(aes(y=predict(lm(mean.size~mean.surv,data=flowclimate_data2))),col="red",linetype=2)+
  geom_point(aes(y=opti.size))+
  geom_line(aes(y=predict(lm(opti.size~mean.surv,data=flowclimate_data2))),linetype=2)+
  theme_bw()+
  labs(title = "red : observed mean size, black : optimal size")+
  ylab("Size at flowering")+
  xlab("Mean probability of surviving")
```

red : observed mean size, black : optimal size



Quand la survie diminue, les plantes qui sont petites survivent moins que les plantes grosses, les petites plantes ne peuvent donc pas fleurir, ce qui diminue la taille à la floraison.

Age at flowering

```
opti_a_year <- NA
for(i in 1:length(annees)){
  opti_a_year[i] <- mean(opti_a_popyear[,i])
}
```

Comparison of optimal age to mean observed age

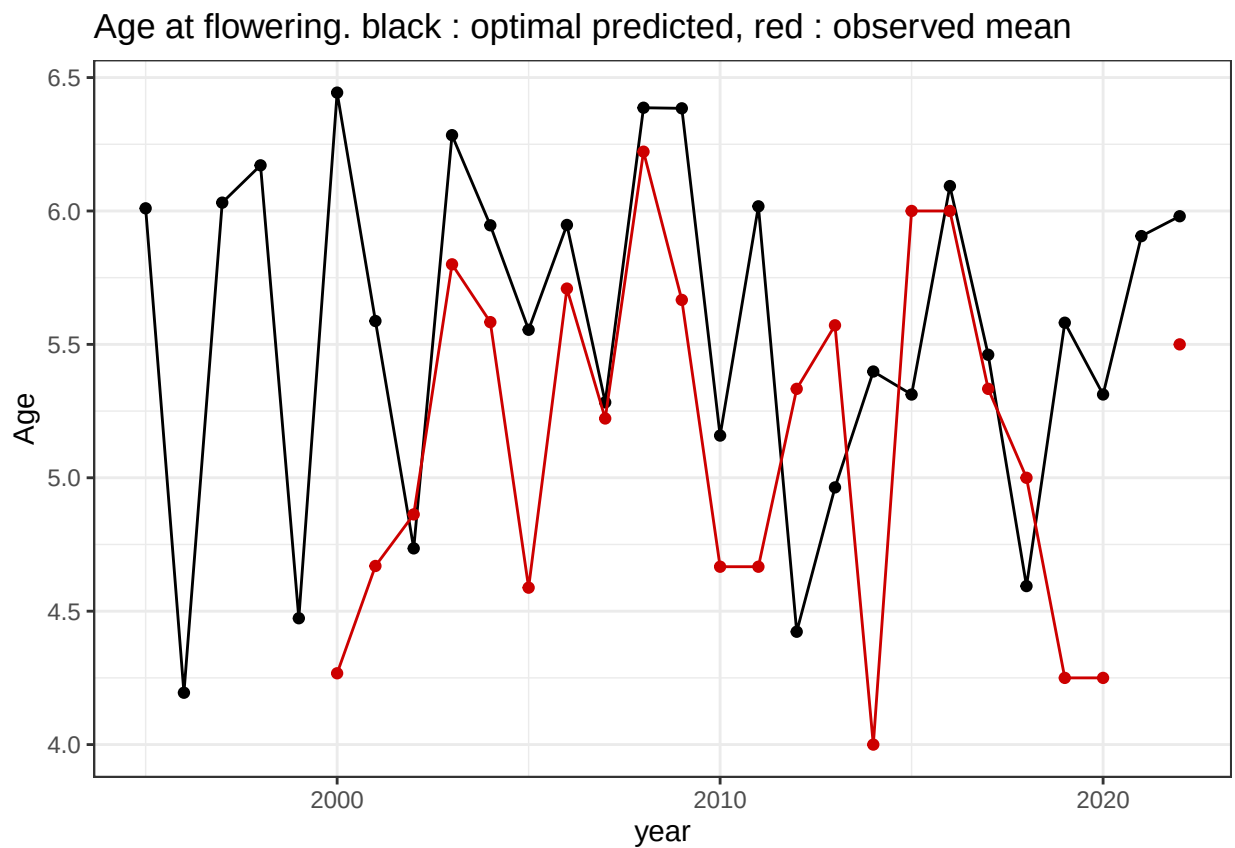
Optimal size at flowering for each year - the model consider that for each year, if we stay in this year indefinitely, the optimal flowering size is opti.size.

```
optimalage_data %>%
  ggplot(aes(x=year)) +
  geom_point(aes(y=opti.age))+
  geom_line(aes(y=opti.age))+
  geom_point(aes(y=mean.age),col="red3")+
  geom_line(aes(y=mean.age),col="red3")+
  # geom_point(aes(y=predict.size),col="blue")+
  # geom_line(aes(y=predict.size),col="blue")+
```

```
theme_bw()+
labs(title="Age at flowering. black : optimal predicted, red : observed mean")+
ylab("Age")
```

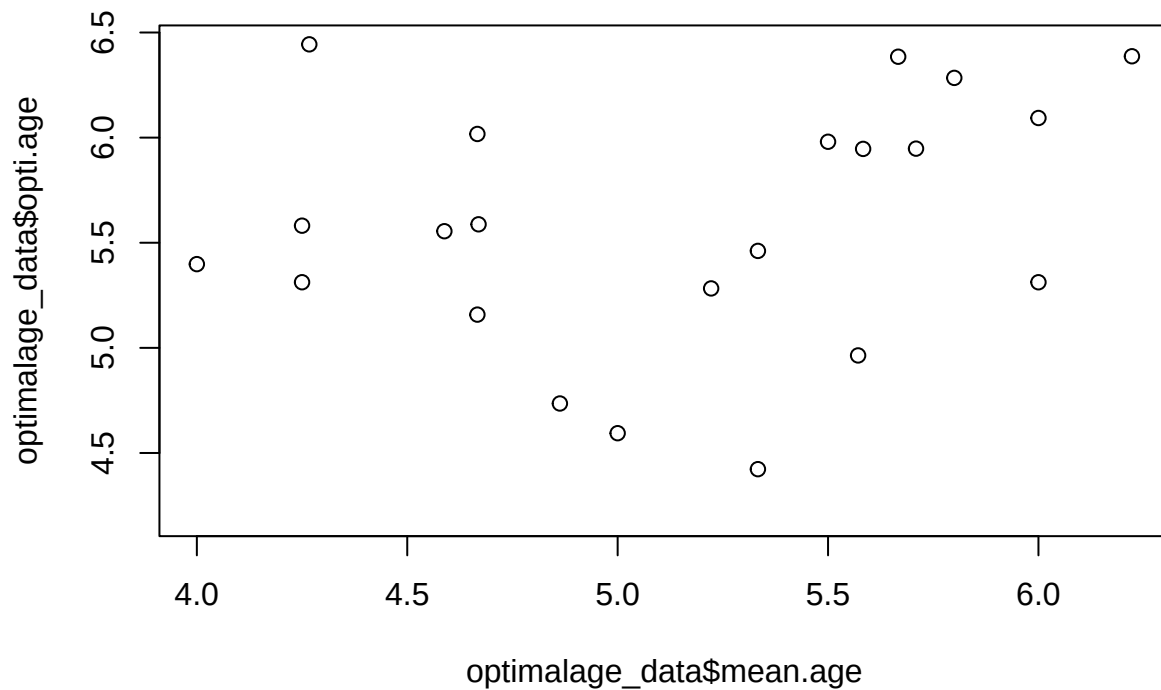
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_line()').
```



Optimal age at flowering compared to observed mean age at flowering

```
plot(x=optimalage_data$mean.age,y=optimalage_data$opti.age)
```



Sensitivity to climatic window

```
seasons <- c("Spring", "Summer", "Autumn", "Winter")

# Sensibilité aux saisons de l'année t
for(s in seasons){
  test_clim <- climate_data_season %>%
    ungroup() %>%
    filter(Season==s,
           year!=1994) %>%
    mutate(opti.age = opti_a_year)

  print(paste(s, "at t, r²=", round(summary(lm(opti.age~Tmean, data=(test_clim)))$r.squared, 5)))
}
```

```
## [1] "Spring at t, r²= 0.00293"
## [1] "Summer at t, r²= 0.0291"
## [1] "Autumn at t, r²= 0.01866"
## [1] "Winter at t, r²= 0.00427"
```

```
# Sensibilité aux saisons de l'année t-1
for(s in seasons){
  test_clim <- climate_data_season %>%
```

```

ungroup() %>%
filter(Season==s,
      year!=2022) %>%
mutate(year = year+1,
      opti.age = opti_a_year)

print(paste(s,"at t-1, r2=", round(summary(lm(opti.age~Tmean,data=(test_clim)))$r.squared,5)))
}

```

```

## [1] "Spring at t-1, r2= 0.00344"
## [1] "Summer at t-1, r2= 0.00198"
## [1] "Autumn at t-1, r2= 0.05065"
## [1] "Winter at t-1, r2= 0.00042"

```

```

test_clim <- climate_data_season %>%
ungroup() %>%
filter(Season %in% c("Autumn","Summer"),
      year!=2022) %>%
group_by(year) %>%
summarise(Tmean = mean(Tmean)) %>%
ungroup() %>%
mutate(year = year+1,
      opti.age = opti_a_year)

print(paste("Autumn and Summer at t-1, r2=", round(summary(lm(opti.age~Tmean,data=(test_clim)))$r.squared,5)))

```

```

## [1] "Autumn and Summer at t-1, r2= 0.03124"

```

```

test_clim <- climate_data_season %>%
ungroup() %>%
filter(Season %in% c("Autumn","Winter"),
      year!=2022) %>%
group_by(year) %>%
summarise(Tmean = mean(Tmean)) %>%
ungroup() %>%
mutate(year = year+1,
      opti.age = opti_a_year)

print(paste("Autumn and Winter at t-1, r2=", round(summary(lm(opti.age~Tmean,data=(test_clim)))$r.squared,5)))

```

```

## [1] "Autumn and Winter at t-1, r2= 0.01479"

```

```

test_clim <- climate_data_season %>%
ungroup() %>%
filter(Season %in% c("Autumn","Winter"),
      year!=2022) %>%
group_by(year) %>%
summarise(Tmean = mean(Tmean)) %>%
ungroup() %>%
mutate(year = year+1,
      opti.age = opti_a_year)

print(paste("Autumn and Winter at t-1, r2=", round(summary(lm(opti.age~Tmean,data=(test_clim)))$r.squared,5)))

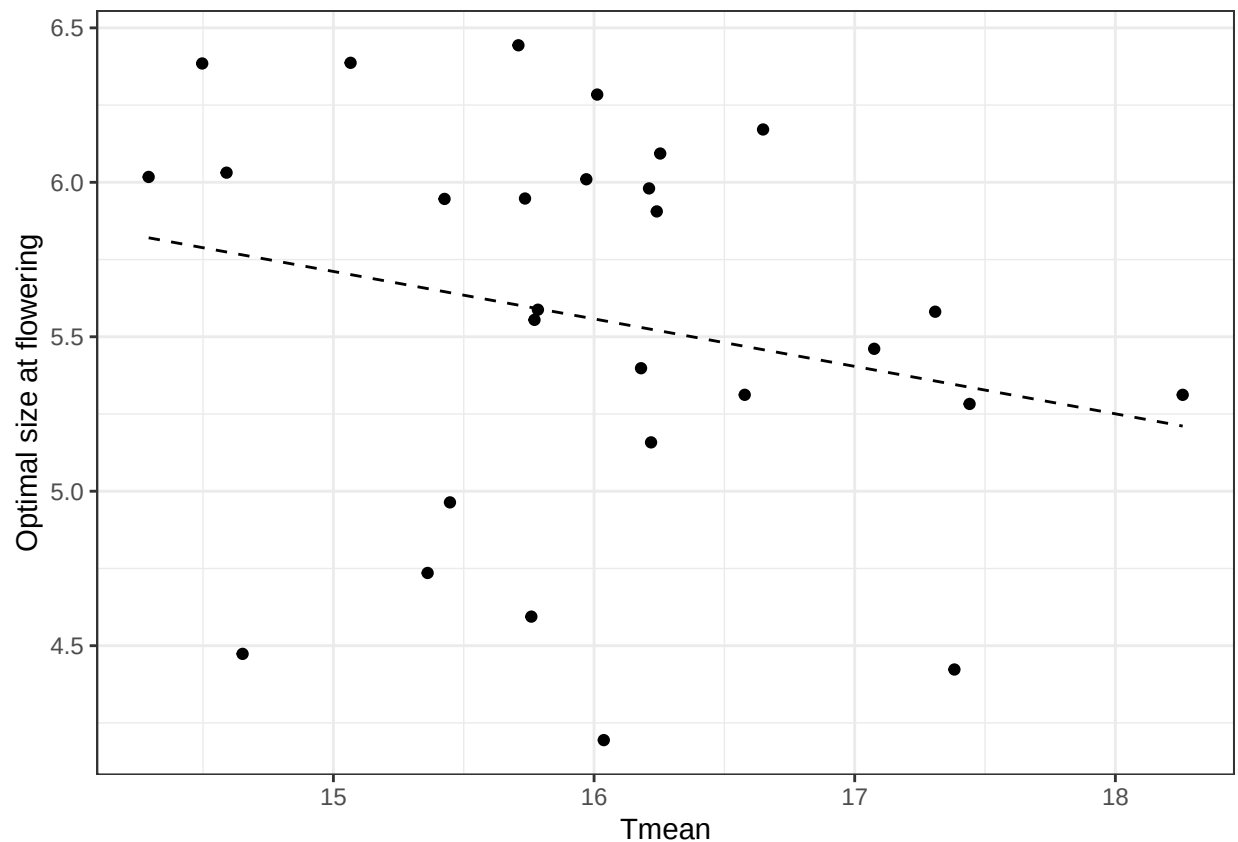
```

```
## [1] "Autumn and Winter at t-1, r²= 0.01479"
```

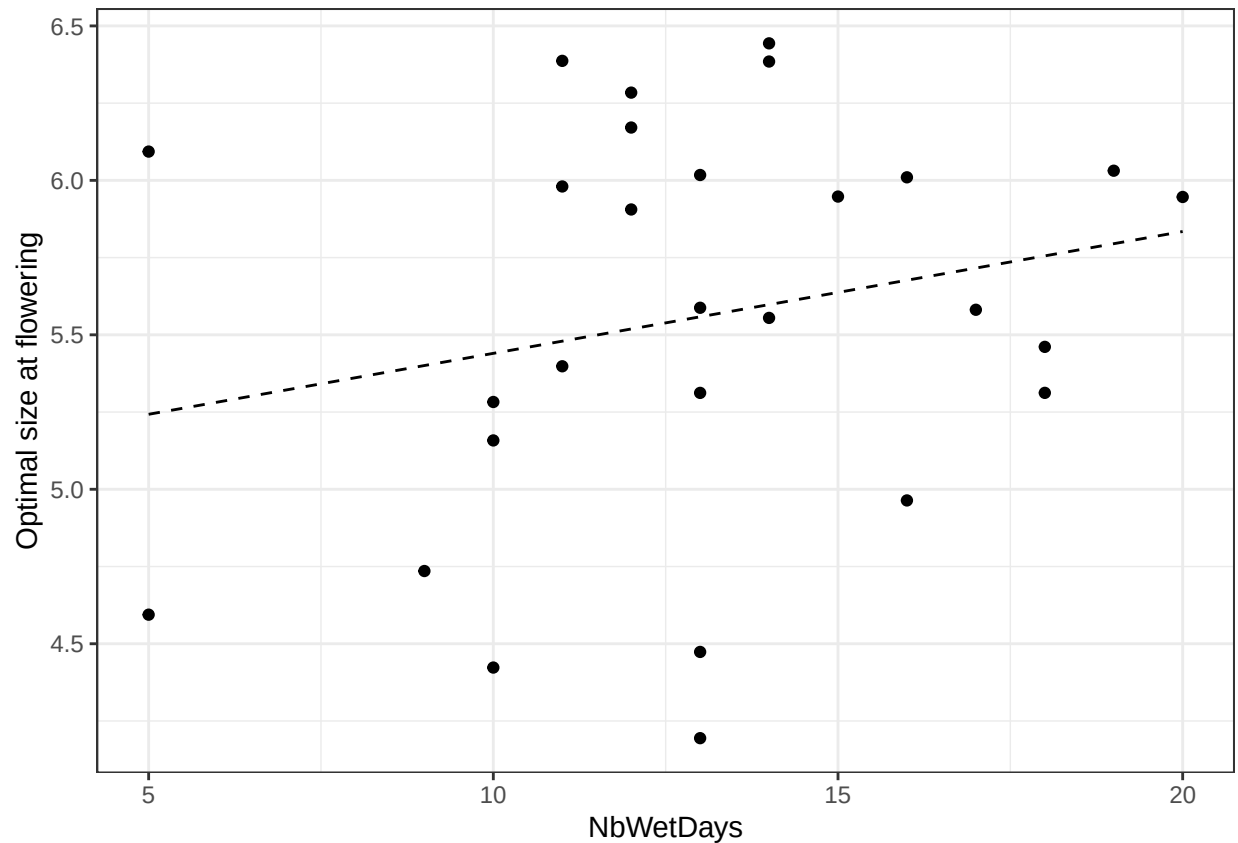
Variations to climatic parameters

```
climate_data_mean_age <- climate_data_season %>%  
  ungroup() %>%  
  filter(Season=="Autumn",  
         year!=2022) %>%  
  mutate(year = year+1,  
         opti.age = opti_a_year)
```

```
climate_data_mean_age %>%  
  ggplot(aes(x=Tmean,y=opti.age)) +  
  geom_point()+  
  geom_line(aes(y=predict(lm(opti.age~Tmean,data=(climate_data_mean_age )))),linetype=2)+  
  theme_bw()+  
  ylab("Optimal size at flowering")
```



```
climate_data_mean_age %>%  
  ggplot(aes(x=NbWetDays,y=opti.age)) +  
  geom_point()+  
  geom_line(aes(y=predict(lm(opti.age~NbWetDays,data=(climate_data_mean_age )))),linetype=2)+  
  theme_bw()+  
  ylab("Optimal size at flowering")
```



```
summary(lm(opti.age~Tmean,data=climate_data_mean_age))$r.squared
```

```
## [1] 0.0506477
```

```
summary(lm(opti.age~NbWetDays,data=climate_data_mean_age ))$r.squared
```

```
## [1] 0.05001674
```

Beta

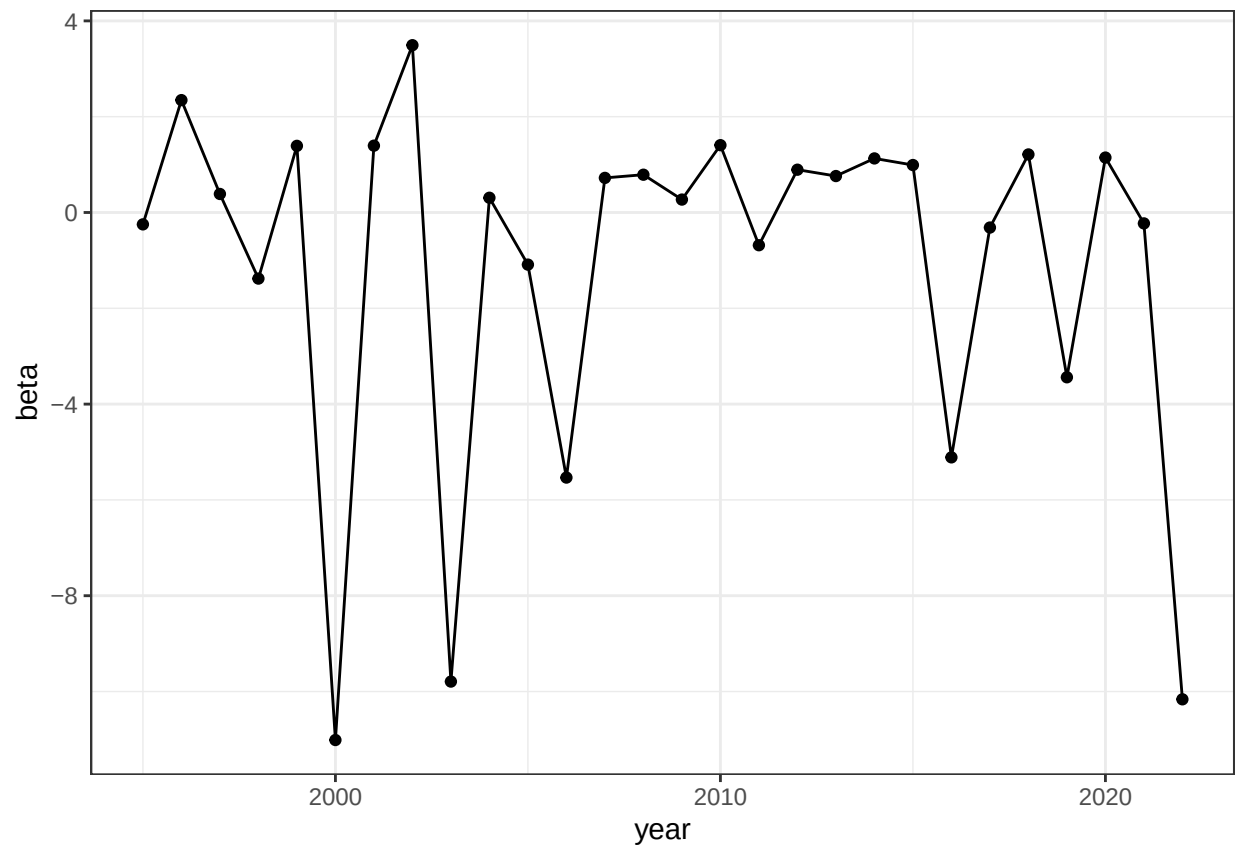
```
load("IPM/Results/opti_beta_popyear.RData")

opti_beta_year <- NA
for(i in 1:length(annees)){
  opti_beta_year[i] <- mean(opti_beta_popyear[,i])
}

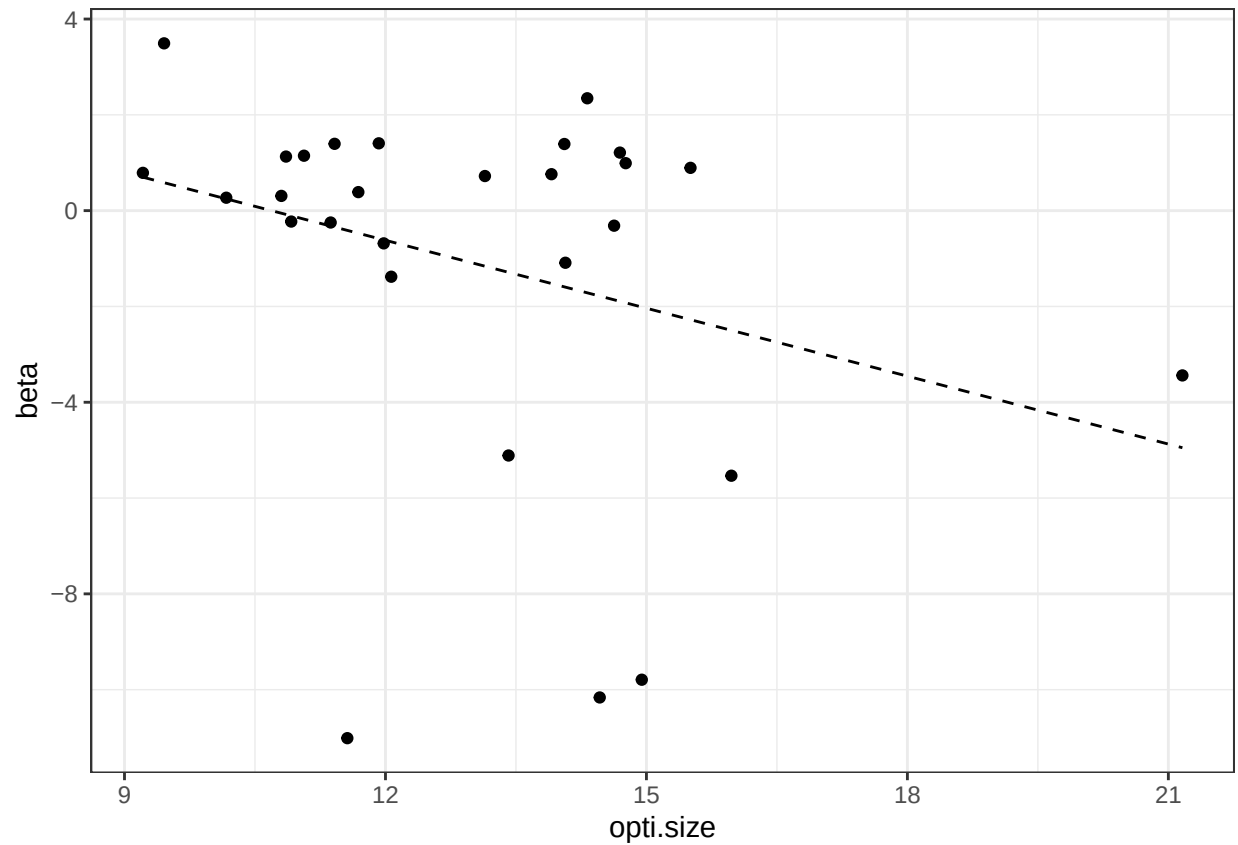
optimalsize_data$beta <- opti_beta_year
```

```
optimalsize_data %>%
  ggplot(aes(y=beta,x=year))+
```

```
geom_point()+
geom_line()+
theme_bw()
```



```
optimalsize_data %>%
  ggplot(aes(y=beta,x=opti.size))+
  geom_point()+
  geom_line(aes(y=predict(lm(beta~opti.size,data=optimalsize_data))),linetype=2)+
  theme_bw()
```



Life history traits compared with climatic data

```
load("IPM/ModelsAIC.RData")
load("IPM/Kernels implementation/IPMKernalYear_AIC.RData")
```

```
annees <- 1995:2022
populations <- c("Po", "Au", "Pe", "E1", "E2", "Cr")
n.size <- 50
minsize <- 0.5
maxsize <- 25
maxsize1 <- 15
h <- (maxsize - minsize) / n.size
h1 <- (maxsize1 - minsize) / n.size
b = minsize + c(0:n.size) * h
#sizes for an individual of age a
#midpoint
ymid.a = 0.5 * (b[1:n.size] + b[2:(n.size + 1)])
#intervals
yI.a <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.a[i,1] <- b[i]
  yI.a[i,2] <- b[i+1]
}
```

```

b1 = minsize + c(0:n.size) * h1
#sizes for an individual of age 1
#midpoint
ymid.1 = 0.5 * (b1[1:n.size] + b1[2:(n.size + 1)])
#intervals
yI.1 <- matrix(nrow = n.size, ncol = 2)
for (i in 1:n.size){
  yI.1[i,1] <- b1[i]
  yI.1[i,2] <- b1[i+1]
}

```

Survival probability

```

#Survival Probability for age 1
sx01 <- function(x,a,Pop,year) {
  fake_data <- expand.grid(
    year = year,
    Pop = Pop,
    Size0Mars = unique(x),
    Age = a)
  predi <- predict(Survglm11, fake_data, allow.new.levels = T)
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean)
  return(meanpredi$V1)
}

#Survival Probability for age > 1
sx02 <- function(x,a,Pop,year) {
  fake_data <- expand.grid(
    year = year,
    Pop = Pop,
    Size0Mars = unique(x),
    Age = a)
  predi <- predict(Survglm12, fake_data, allow.new.levels = T)
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean)
  return(meanpredi$V1)
}

```

```

mean.surv <- NULL
yr <- 1
for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
  W <- function(age) {
    return(N[(50 * (age - 1) + 1):(50 * age)])
  }
  mean.surv[yr] <- sum(sx01(ymid.1,1,Pop=populations,year=annees[yr])*W(1)/sum(W(1)))*sum(W(1))

  for (a in 2:8) {

```

```

    mean.surv[yr] <- mean.surv[yr] + sum(sx02(ymid.a,a,Pop=populations,year=annees[yr])*W(a)/sum(W(a)))
  }
}

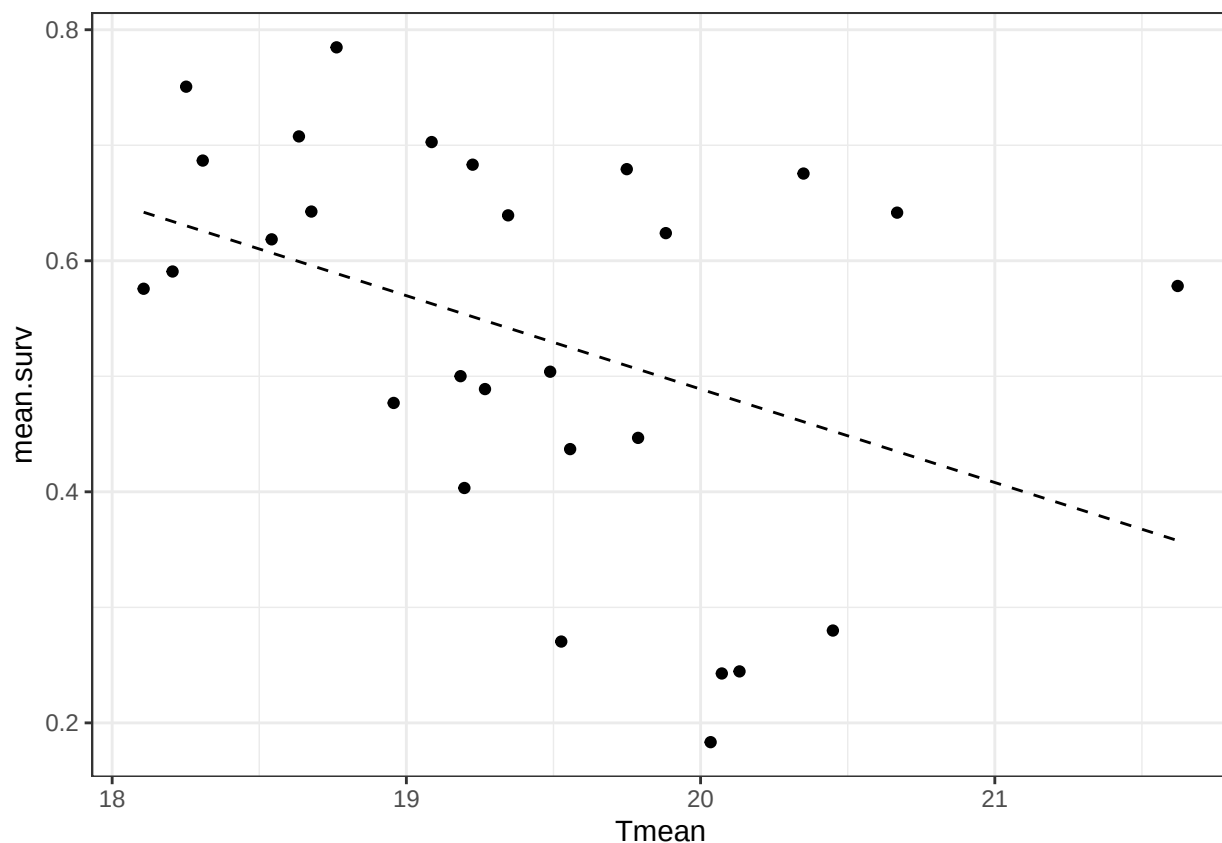
survpredi_data <- tibble(year=annees,
  mean.surv=mean.surv,
  Tmean=climate_data_mean_lambda$Tmean,
  NbWetDays=climate_data_mean_lambda$NbWetDays)

```

```

survpredi_data %>%
  ggplot(aes(x = Tmean, y = mean.surv))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.surv~Tmean,data=survpredi_data))),linetype=2)+
  theme_bw()

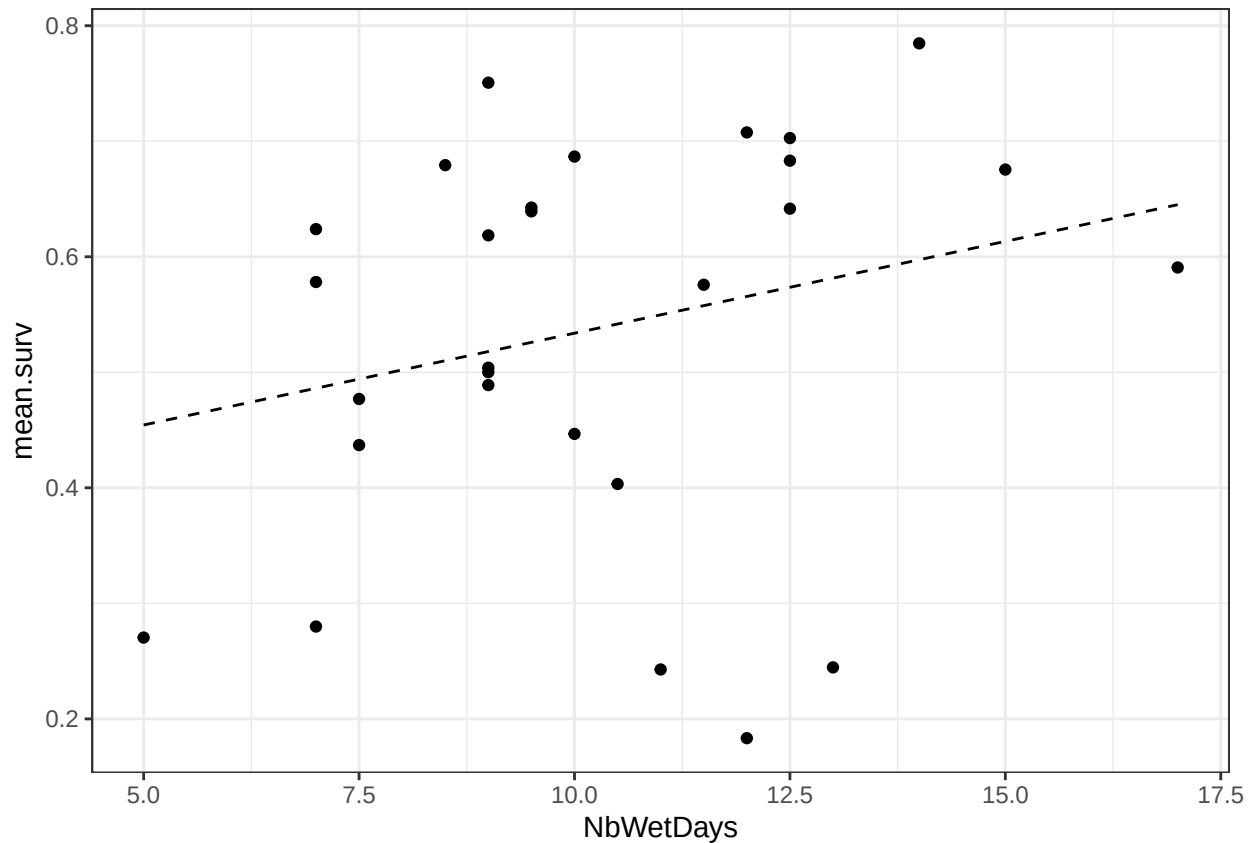
```



```

survpredi_data %>%
  ggplot(aes(x = NbWetDays, y = mean.surv))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.surv~NbWetDays,data=survpredi_data))),linetype=2)+
  theme_bw()

```



The mean survival probability is positively correlated with the temperature. A plant has less chance to survive if it's warmer, ok.

Flowering probability

```
f1r0 <- function(x, a, Beta=0,Pop,year) {
  fake_data <- expand.grid(
    year = year,
    Pop = Pop,
    Size0Mars = unique(x),
    Age = a)
  predi <- plogis(predict(Flowglm1, newdata = fake_data, allow.new.levels = T, type = "link") + Beta)
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean)
  return(meanpredi$V1)
}
```

```
mean.flow <- NULL
yr <- 1
for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
}
```

```

W <- function(age) {
  return(N[(50 * (age - 1) + 1):(50 * age)])
}
mean.flow[yr] <- sum(flR0(yMid.1,1,Pop=populations,year=annees[yr])*W(1)/sum(W(1)))*sum(W(1))

for (a in 2:8) {
  mean.flow[yr] <- mean.flow[yr] + sum(flR0(yMid.a,a,Pop=populations,year=annees[yr])*W(a)/sum(W(a)))
}
}

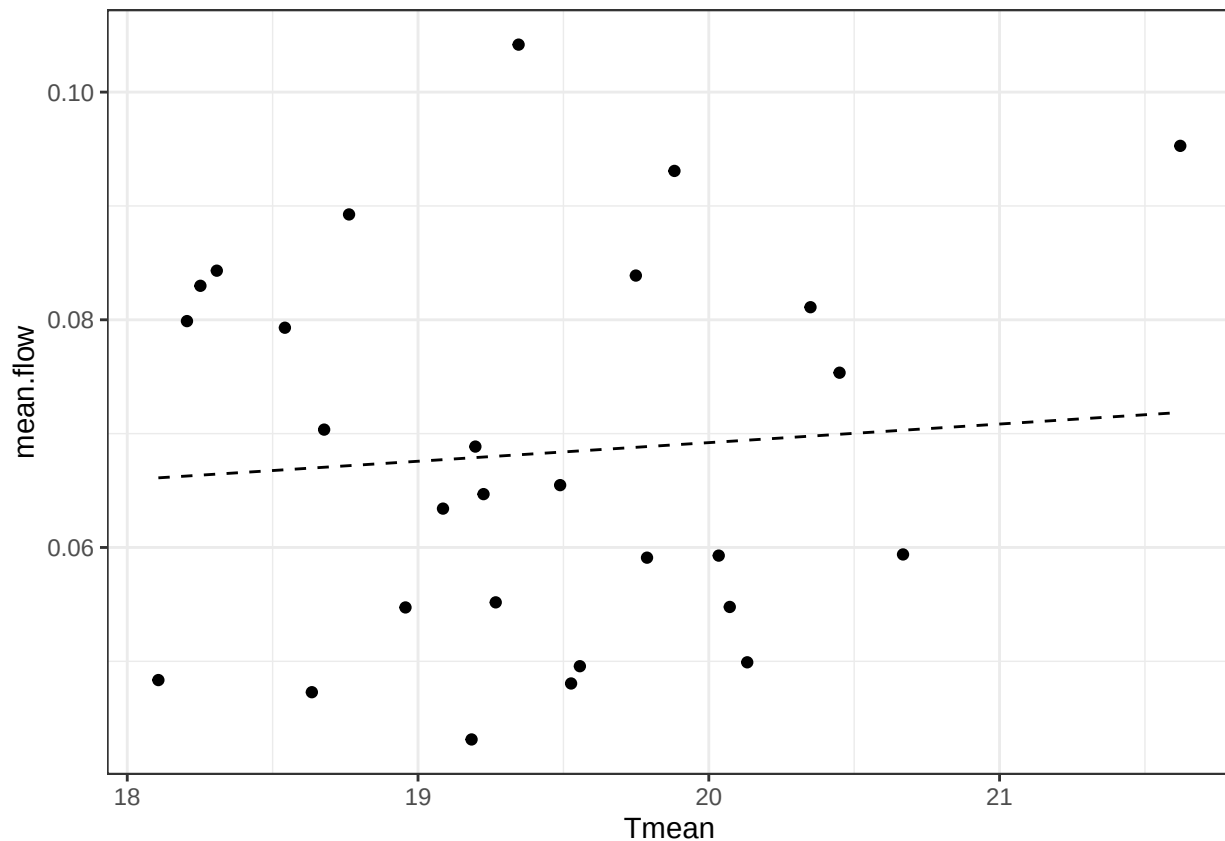
flowpredi_data <- tibble(year=annees,
  mean.flow=mean.flow,
  Tmean=climate_data_mean_lambda$Tmean,
  NbWetDays=climate_data_mean_lambda$NbWetDays)

```

```

flowpredi_data %>%
  ggplot(aes(x = Tmean, y = mean.flow))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.flow~Tmean,data=flowpredi_data))),linetype=2)+
  theme_bw()

```

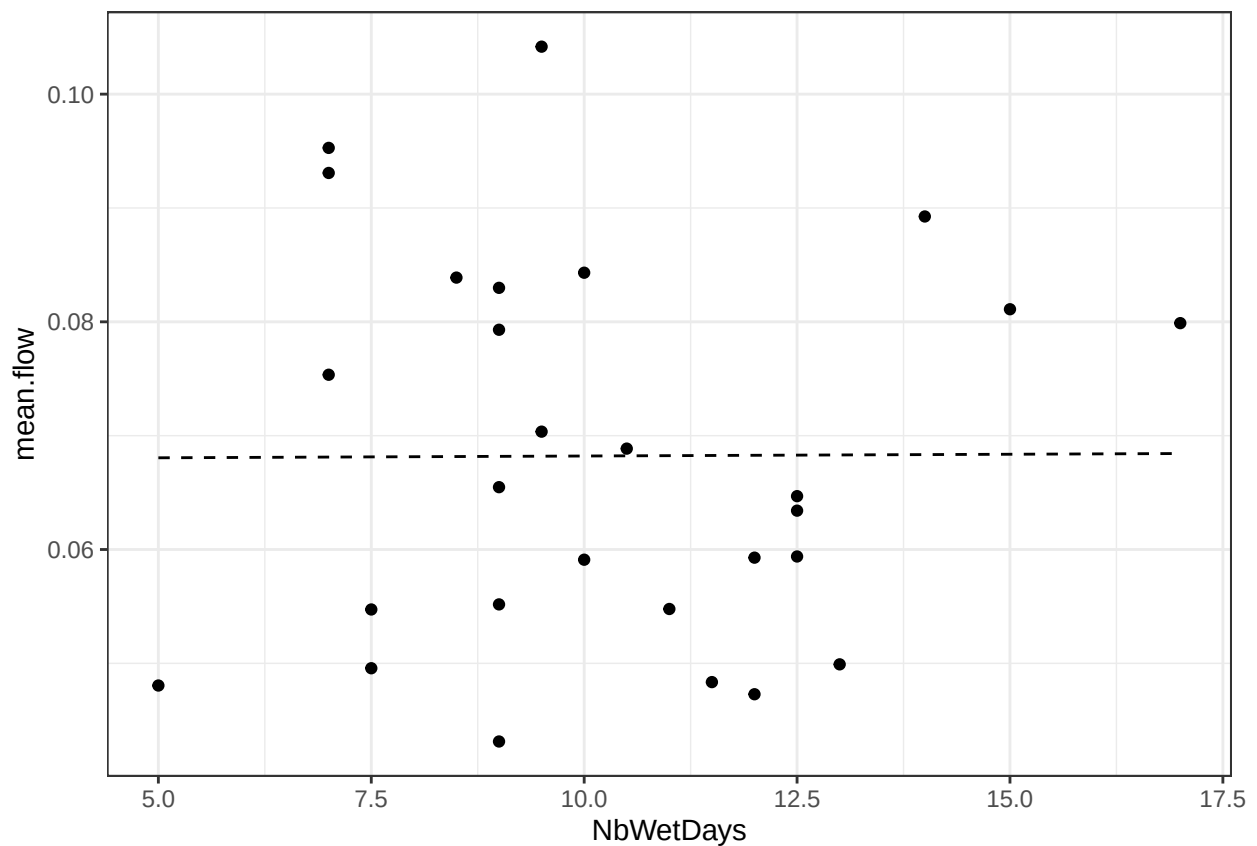


```

flowpredi_data %>%
  ggplot(aes(x = NbWetDays, y = mean.flow))+
  geom_point()+

```

```
geom_line(aes(y=predict(lm(mean.flow~NbWetDays,data=flowpredi_data))),linetype=2)+
theme_bw()
```



The mean flowering probability is positively correlated with temperature. If the year is warmer, a plant will flower now instead of wait the next year and risk to die during summer.

Growth

```
Gyx3 <- function(ysup,yinf,x,a,Pop,year){
  sizepow <- 0.4343354
  yinfpow <- yinf**sizepow
  ysuppow <- ysup**sizepow
  yinfpow[1] <- -Inf
  ysuppow[length(ysuppow)] <- Inf

  fake_data <- expand.grid(
    year = year,
    Pop = Pop,
    Size0Mars = unique(x),
    Age = a)
  predi <- predict(Growthglm2, fake_data, allow.new.levels = T) #Prédit les taille^pow à t+1 possibles
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean) #Calcule la moyenne des tallet+1

  resvar <- predict(Growthglm2$resid_fit, newdata = fake_data, allow.new.levels = T)
```

```

meanresvar <- aggregate(resvar[,1], list(fake_data$SizeOMars), mean)$x
cdf.sup <- mapply(pnorm, mean = meanpredi$V1, sd = sqrt(meanresvar), MoreArgs = list(q=ysuppow))
cdf.inf <- mapply(pnorm, mean = meanpredi$V1, sd = sqrt(meanresvar), MoreArgs = list(q=yinfpow))
return(cdf.sup - cdf.inf)
}

```

```

mean.size <- NULL
mean.size1 <- array(0,dim=length(annees))

for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
  W <- function(age) {
    return(N[(50 * (age - 1) + 1):(50 * age)])
  }

  G <- Gyx3(ysup=yI.a[,2], yinf=yI.a[,1], x=yid.1, a=1, Pop=populations, year=annees[yr])
  for(i in 1:50){
    mean.size1[yr] <- mean.size1[yr] + yid.a[i]*sum(G[i,]*W(1))
  }
  mean.size[yr] <- sum(yid.1*W(1)/sum(W(1))*sum(W(1)))

  for (a in 2:8) {
    mean.size[yr] <- mean.size[yr] + sum(yid.a*W(a)/sum(W(a))) * sum(W(a))
    G <- Gyx3(ysup=yI.a[,2], yinf=yI.a[,1], x=yid.a, a=a, Pop=populations, year=annees[yr])
    for(i in 1:50){
      mean.size1[yr] <- mean.size1[yr] + yid.a[i]*sum(G[i,]*W(a))
    }
  }
}

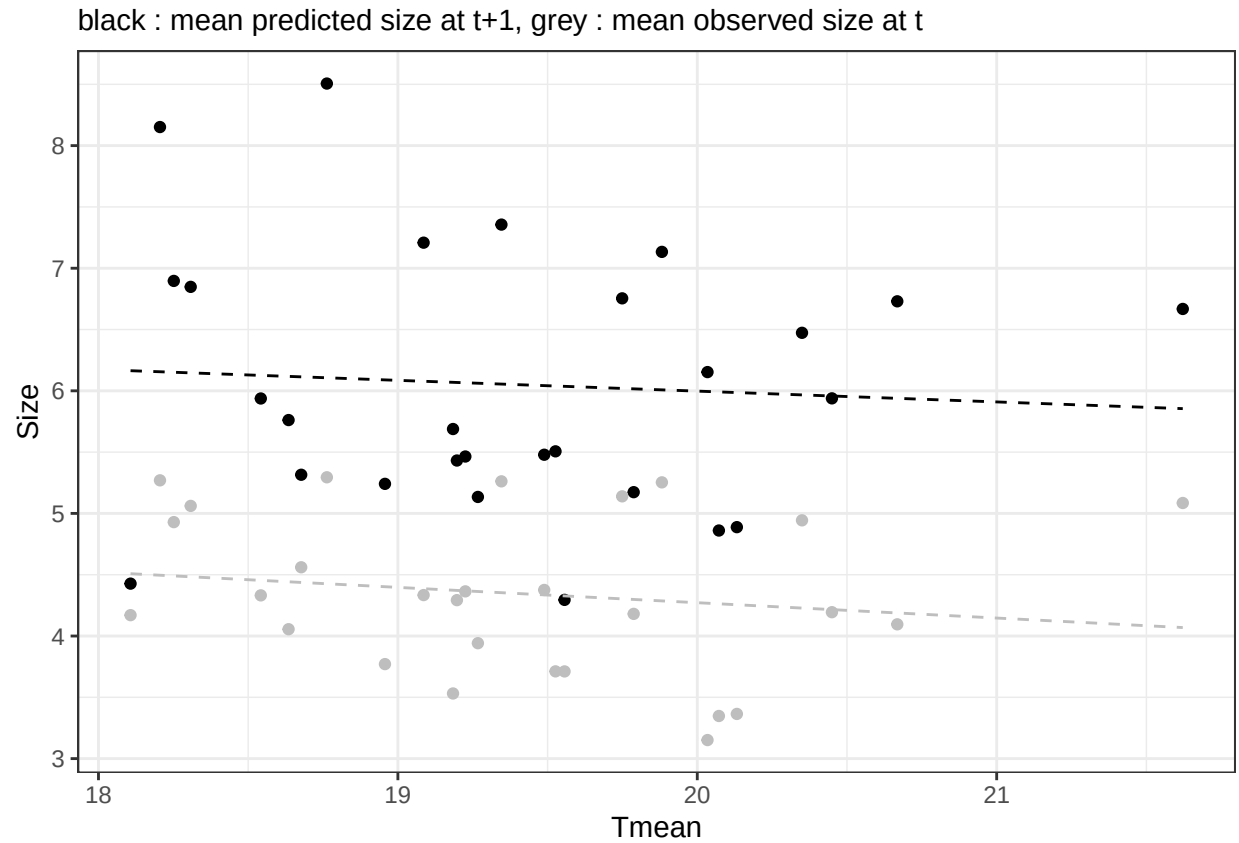
growthpredi_data <- tibble(year=annees,
                           mean.size=mean.size,
                           mean.size1=mean.size1,
                           Tmean=climate_data_mean_lambda$Tmean,
                           NbWetDays=climate_data_mean_lambda$NbWetDays) %>%
  mutate(grow.rate = mean.size1/mean.size)

```

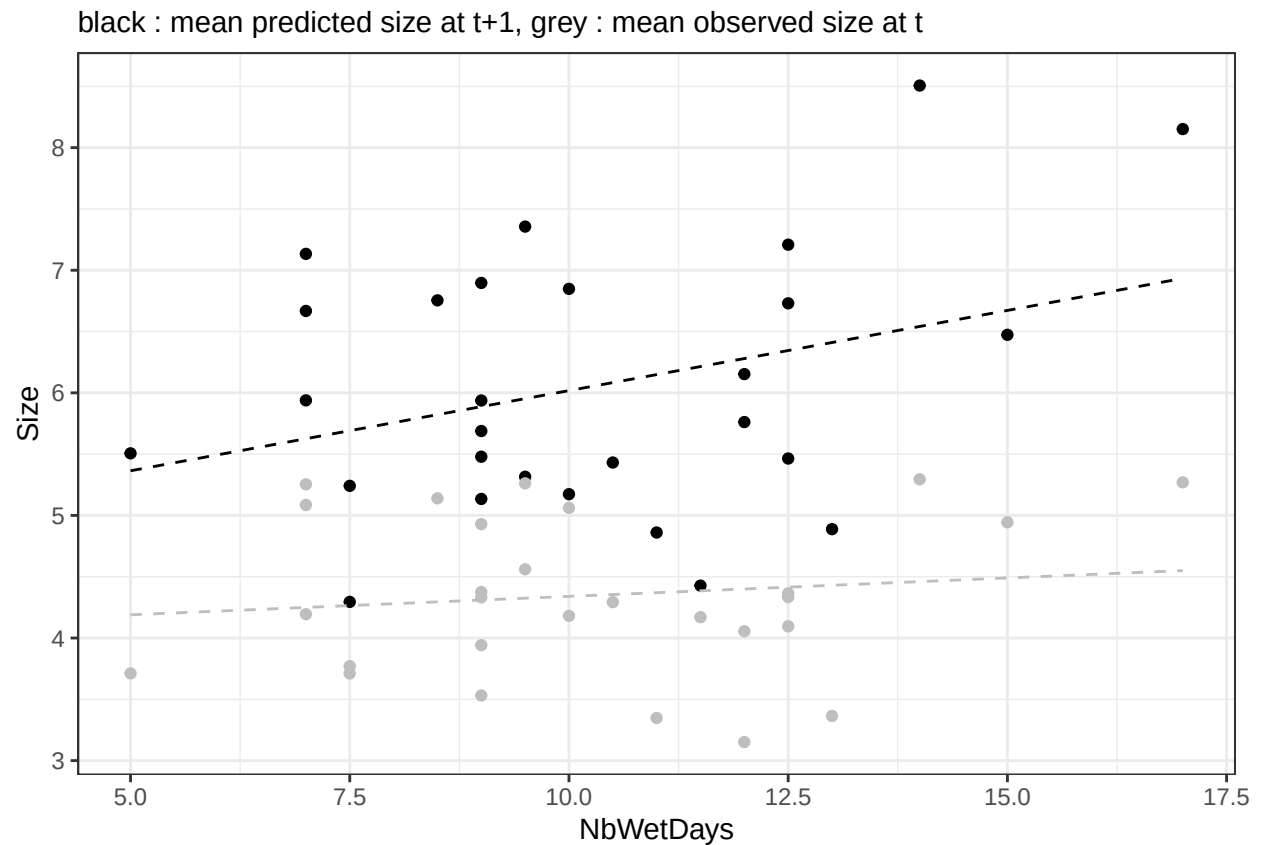
```

growthpredi_data %>%
  ggplot(aes(x = Tmean))+
  geom_point(aes(y = mean.size1))+
  geom_line(aes(y=predict(lm(mean.size1~Tmean,data=growthpredi_data))),linetype=2)+
  geom_point(aes(y = mean.size),col="grey")+
  geom_line(aes(y=predict(lm(mean.size~Tmean,data=growthpredi_data))),linetype=2,col="grey")+
  theme_bw()+
  labs(subtitle = "black : mean predicted size at t+1, grey : mean observed size at t")+
  ylab("Size")

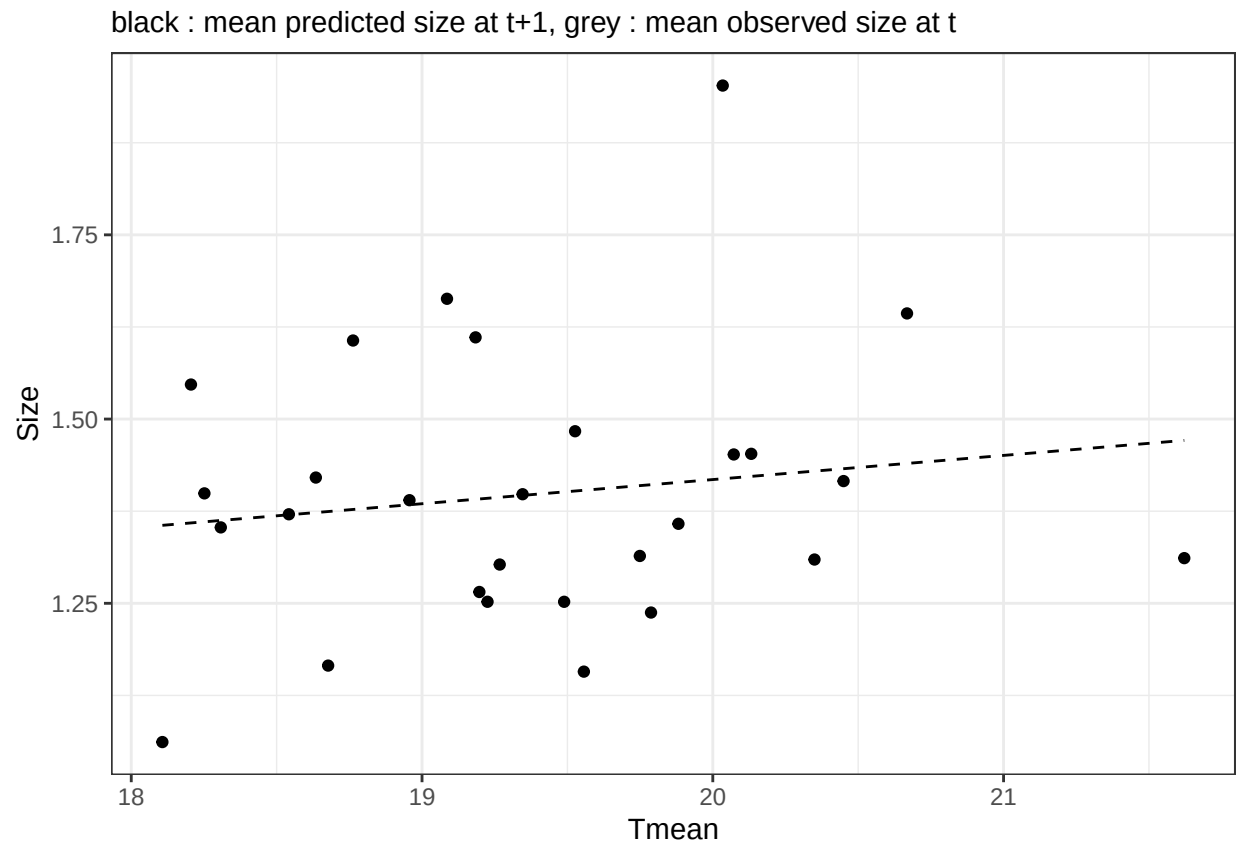
```



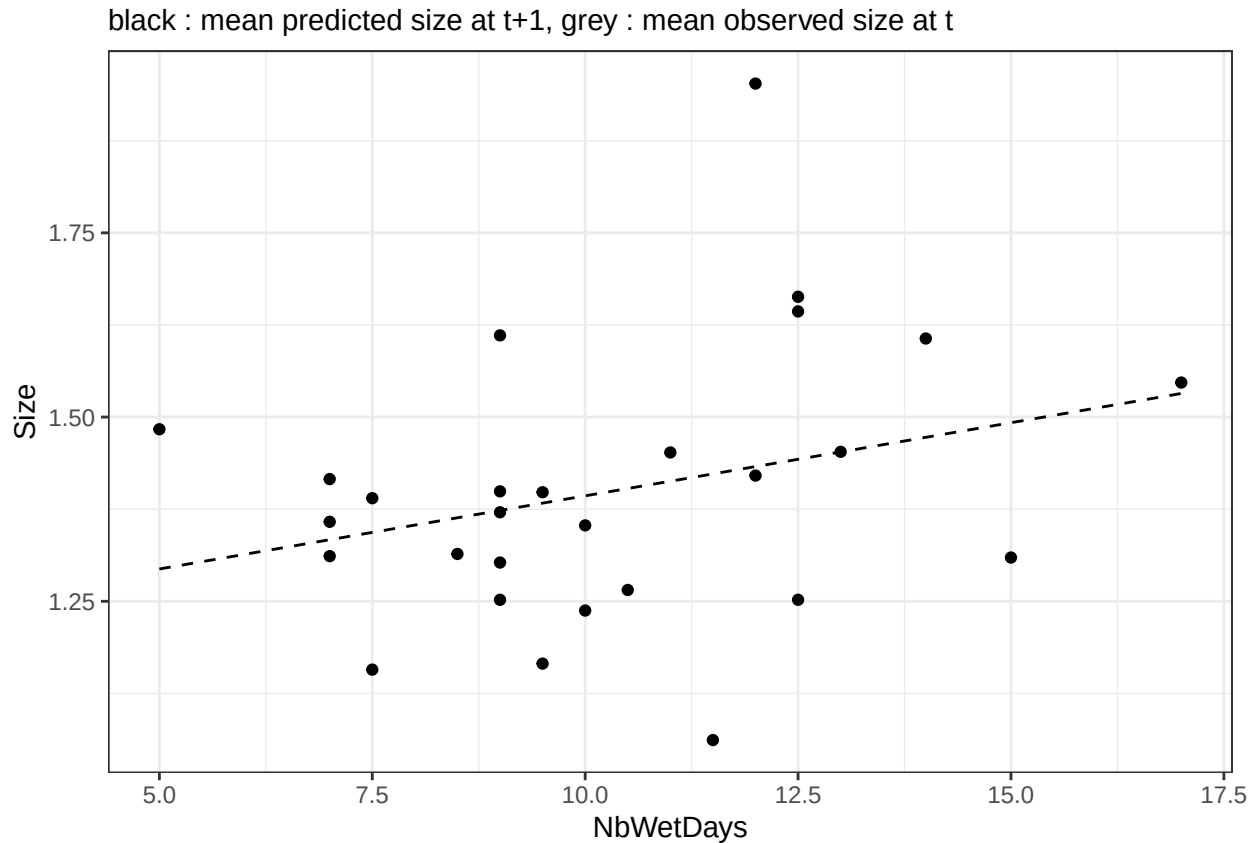
```
growthpredi_data %>%
  ggplot(aes(x = NbWetDays))+
  geom_point(aes(y = mean.size1))+
  geom_line(aes(y=predict(lm(mean.size1~NbWetDays,data=growthpredi_data))),linetype=2)+
  geom_point(aes(y = mean.size),col="grey")+
  geom_line(aes(y=predict(lm(mean.size~NbWetDays,data=growthpredi_data))),linetype=2,col="grey")+
  theme_bw()+
  labs(subtitle = "black : mean predicted size at t+1, grey : mean observed size at t")+
  ylab("Size")
```



```
growthpredi_data %>%
  ggplot(aes(x = Tmean))+
  geom_point(aes(y = grow.rate))+
  geom_line(aes(y=predict(lm(grow.rate~Tmean,data=growthpredi_data))),linetype=2)+
  theme_bw()+
  labs(subtitle = "black : mean predicted size at t+1, grey : mean observed size at t")+
  ylab("Size")
```



```
growthpredi_data %>%
  ggplot(aes(x = NbWetDays))+
  geom_point(aes(y = grow.rate))+
  geom_line(aes(y=predict(lm(grow.rate~NbWetDays,data=growthpredi_data))),linetype=2)+
  theme_bw()+
  labs(subtitle = "black : mean predicted size at t+1, grey : mean observed size at t")+
  ylab("Size")
```



Fecundity

```
numbofcap <- function(x, a, year) {
  fake_data <- expand.grid(
    year = year,
    Size0Mars = unique(x),
    Age = a)
  predi <- exp(predict(Cptlglm1, fake_data)) # prédiction du nombre de capitules produits pour iun indi
  meanpredi <- aggregate(predi, list(fake_data$Size0Mars), mean) #transforme le vecteur de prédiction e
  nbcap <- ifelse(meanpredi$V1>=0,meanpredi$V1,0)
  return(matrix(nbcap, length(x), length(x), byrow = T))
}
```

```
mean.capi <- NULL
yr <- 1
for(yr in 1:length(annees)) {
  A <- eigen.analysis(IPMKernalYear_AIC[[yr]])
  N <- matrix(ncol = 8, nrow = 50)
  for (i in 1:8) {
    N[, i] <- A$stable.stage[(50 * (i - 1) + 1):(50 * i)]
  }
  W <- function(age) {
    return(N[(50 * (age - 1) + 1):(50 * age)])
  }
}
```

```

mean.capi[yr] <- sum(flr0(ymid.1,1,Pop=populations,year=annees[yr])*numbofcap(ymid.1,1,year=annees[yr])

for (a in 2:8) {
  mean.capi[yr] <- mean.capi[yr] + sum(flr0(ymid.a,a,Pop=populations,year=annees[yr])*numbofcap(ymid.a,a,year=annees[yr])
}
}

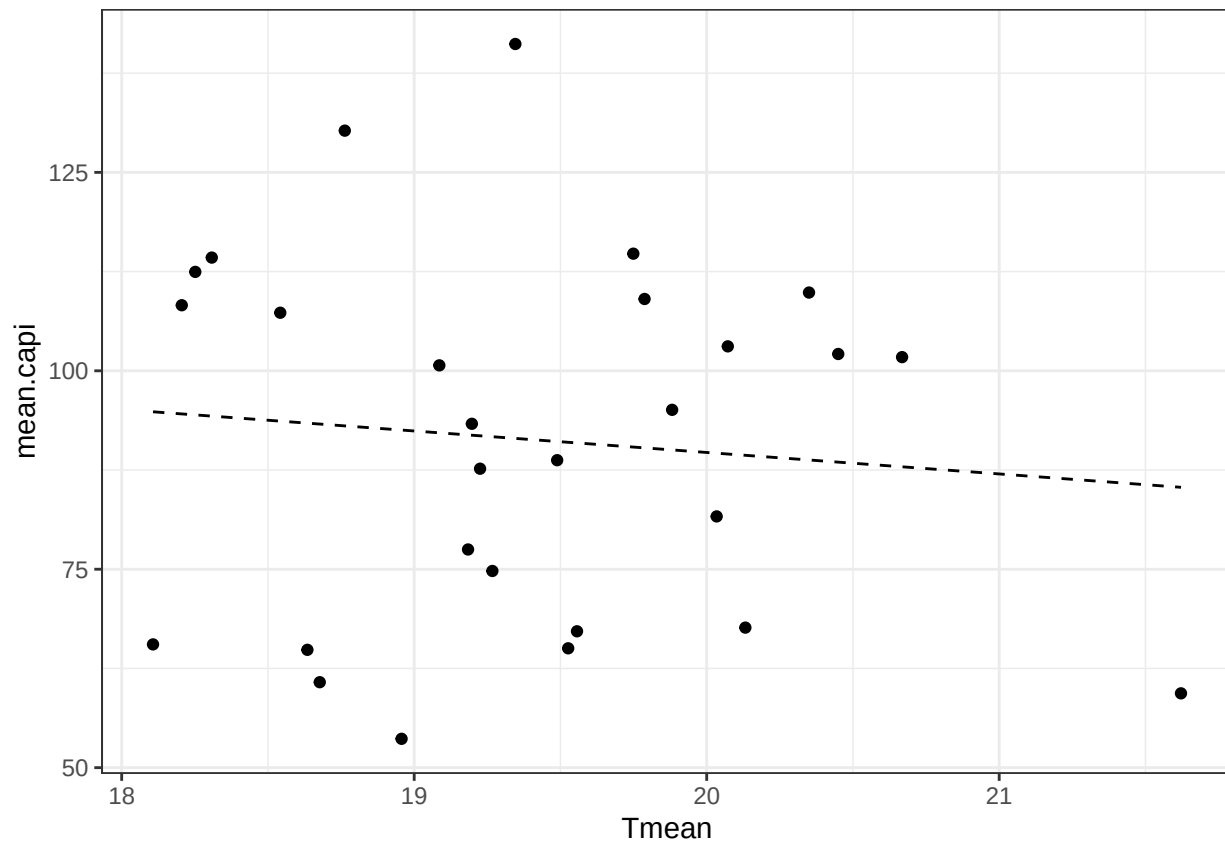
capipredi_data <- tibble(year=annees,
  mean.capi=mean.capi,
  Tmean=climate_data_mean_lambda$Tmean,
  NbWetDays=climate_data_mean_lambda$NbWetDays)

```

```

capipredi_data %>%
  ggplot(aes(x = Tmean, y = mean.capi))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.capi~Tmean,data=capipredi_data))),linetype=2)+
  theme_bw()

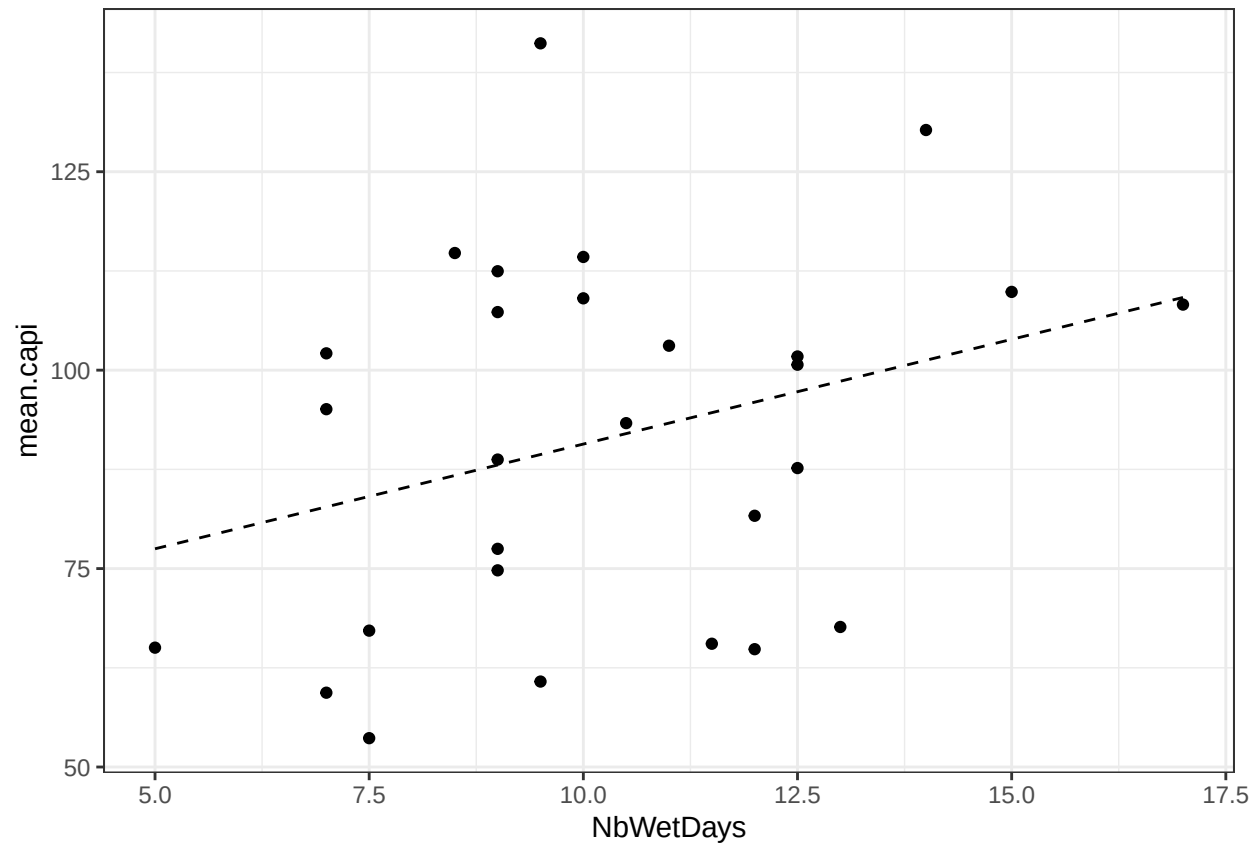
```



```

capipredi_data %>%
  ggplot(aes(x = NbWetDays, y = mean.capi))+
  geom_point()+
  geom_line(aes(y=predict(lm(mean.capi~NbWetDays,data=capipredi_data))),linetype=2)+
  theme_bw()

```

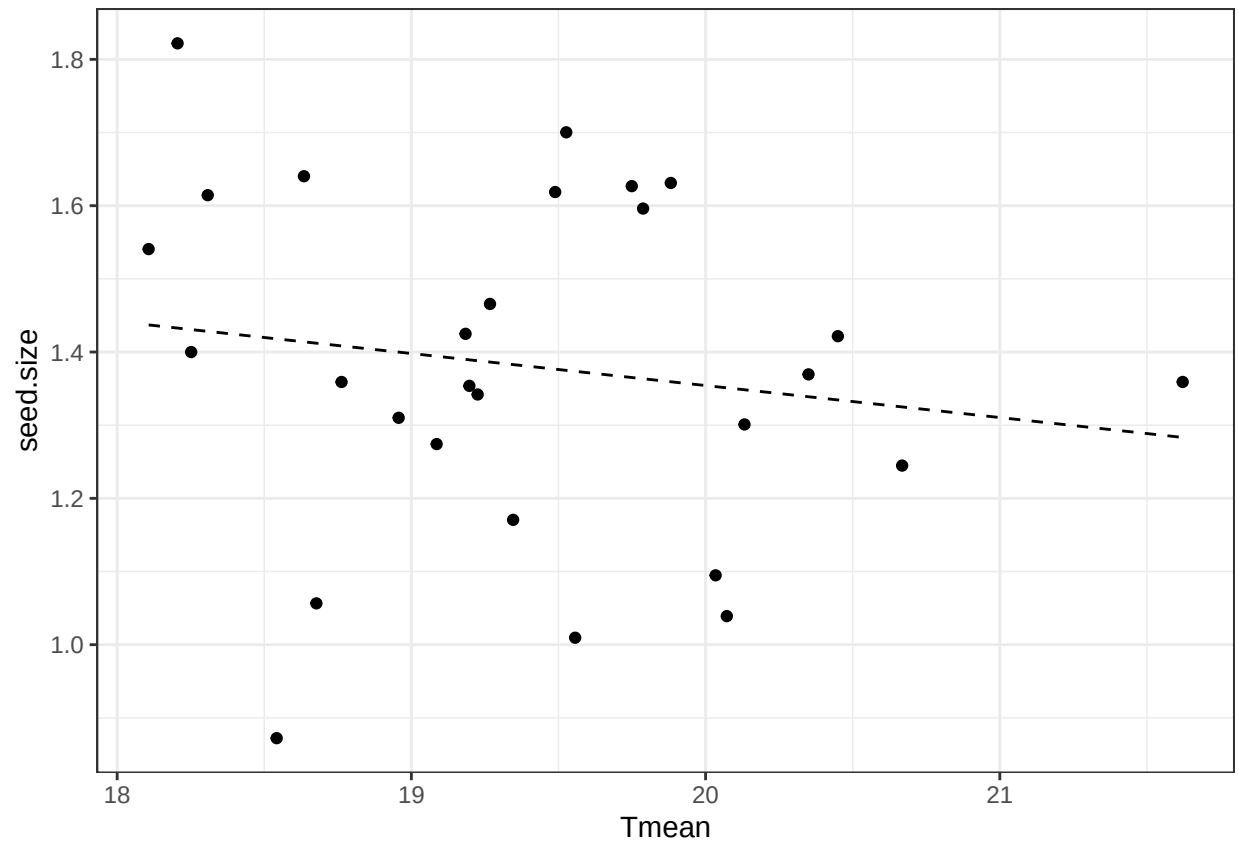


Seedling size

Effet de la température à t sur la taille des seedlings à $t+1$

```
seed.size <- NULL
for(yr in 1:length(annees)){
  seed.size[yr] <- mean(predict(Pltglm1, newdata = expand.grid(year=annees[yr],Pop=populations))[,1])
}
seedsizepredi_data <- tibble(year=annees,
                             Tmean=climate_data_mean_lambda$Tmean,
                             NbWetDays=climate_data_mean_lambda$NbWetDays,
                             seed.size=seed.size)
```

```
seedsizepredi_data %>%
  ggplot(aes(x = Tmean, y = seed.size))+
  geom_point()+
  geom_line(aes(y=predict(lm(seed.size~Tmean,data=seedsizepredi_data))),linetype=2)+
  theme_bw()
```



```
seedsizepredi_data %>%  
  ggplot(aes(x = NbWetDays, y = seed.size))+  
  geom_point()+  
  geom_line(aes(y=predict(lm(seed.size~NbWetDays,data=seedsizepredi_data))),linetype=2)+  
  theme_bw()
```

